

6 SYMMETRIES & CONSERVATION LAWS

6.1 INTRODUCTION

Conservation laws (energy, momentum, and angular momentum) are familiar from your first course in *classical* mechanics. These same conservation laws hold in quantum mechanics; in both contexts they are the result of symmetries. In this chapter we will explain what a symmetry is and what it means for something to be conserved in quantum mechanics—and show how the two are related. Along the way we’ll investigate two related properties of quantum systems—energy level degeneracy and the selection rules that distinguish allowed from “forbidden” transitions.

What is a symmetry? It is some transformation that leaves the system unchanged. As an example consider rotating a square piece of paper, as shown in Figure 6.1. If you rotate it by 30° about an axis through its center it will be in a different orientation than the one it started in, but if you rotate it by 90° it will resume its original orientation; you wouldn’t even know it had been rotated unless (say) you wrote numbers on the corners (in which case they would be permuted). A square therefore has a *discrete* rotational symmetry: a rotation by $n\pi/2$ for any integer n leaves it unchanged.¹ If you repeated this experiment with a circular piece of paper, a rotation by *any* angle would leave it unchanged; the circle has *continuous* rotational symmetry. We will see that both discrete and continuous symmetries are important in quantum mechanics.

Now imagine that the shapes in Figure 6.1 refer not to pieces of paper, but to the boundaries of a two-dimensional infinite square well. In that case the potential energy would have the same rotational symmetries as the piece of paper and (because the kinetic energy is unchanged by a rotation) the Hamiltonian would also be invariant. In quantum mechanics, when we say that a system has a symmetry, this is what we mean: that the Hamiltonian is unchanged by some transformation, such as a rotation or a translation.

6.1.1 Transformations in Space

In this section, we introduce the quantum mechanical operators that implement translations, inversions, and rotations. We define each of these operators by how it acts on an arbitrary function. The **translation operator** takes a function and shifts it a distance a . The operator that accomplishes this is defined by the relation

¹ A square of course has other symmetries as well, namely mirror symmetries about axes along a diagonal or bisecting two sides. The set of all transformations that leave the square unchanged is called D_4 , the “dihedral group” of degree 4.

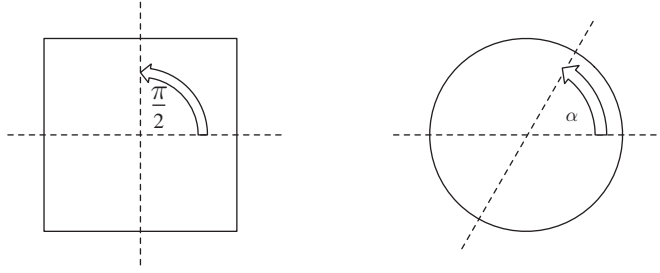


Figure 6.1: A square has a discrete rotational symmetry; it is unchanged when rotated by $\pi/2$ or multiples thereof. A circle has continuous rotational symmetry; it is unchanged when rotated by any angle α .

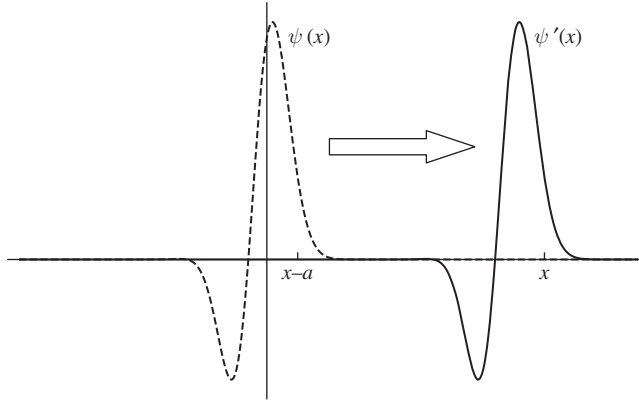


Figure 6.2: A wave function $\psi(x)$ and the translated wave function $\psi'(x) = \hat{T}(a)\psi(x)$. Note that the value of ψ' at x is equal to the value of ψ at $x - a$.

$$\hat{T}(a)\psi(x) = \psi'(x) = \psi(x - a). \quad (6.1)$$

The sign can be confusing at first; this equation says that the translated function ψ' at x is equal to the untranslated function ψ at $x - a$ (Figure 6.2)—the function itself has been shifted to the *right* by an amount a .

The operator that reflects a function about the origin, the **parity operator** in one dimension, is defined by

$$\hat{\Pi}\psi(x) = \psi'(x) = \psi(-x).$$

The effect of parity is shown graphically in Figure 6.3. In three dimensions parity changes the sign of all three coordinates: $\hat{\Pi}\psi(x, y, z) = \psi(-x, -y, -z)$.²

Finally, the operator that rotates a function about the z axis through an angle φ is most naturally expressed in polar coordinates as

$$\hat{R}_z(\varphi)\psi(r, \theta, \phi) = \psi'(r, \theta, \phi) = \psi(r, \theta, \phi - \varphi). \quad (6.2)$$

² The parity operation in three dimensions can be realized as a mirror reflection followed by a rotation (see Problem 6.1). In two dimensions, the transformation $\psi'(x, y) = \psi(-x, -y)$ is no different from a 180° rotation. We will use the term parity exclusively for spatial inversion, $\hat{\Pi}\psi(\mathbf{r}) = \psi(-\mathbf{r})$, in one or three dimensions.

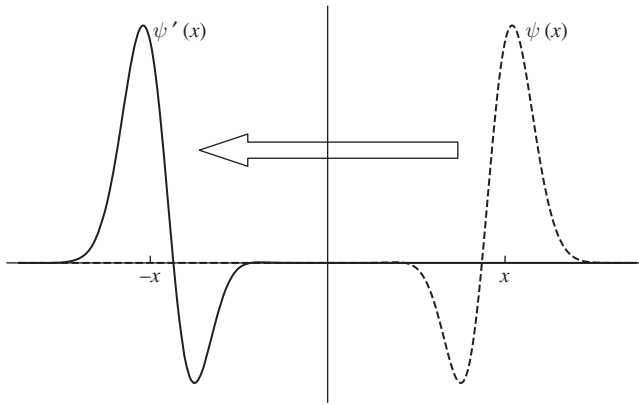


Figure 6.3: A function $\psi(x)$ and the function $\psi'(x) = \hat{\Pi} \psi(x) = \psi(-x)$ after a spatial inversion. The value of ψ' at x is equal to the value of ψ at $-x$.

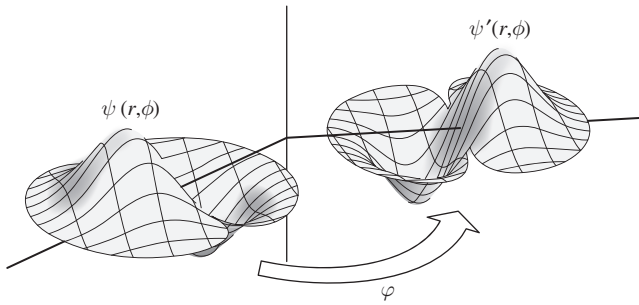


Figure 6.4: A function $\psi(r, \phi)$ and the rotated function $\psi'(r, \phi) = \psi(r, \phi - \varphi)$ after a counter-clockwise rotation about the vertical axis by an angle φ .

When we take up the study of rotations in Section 6.5, we will introduce expressions for rotations about arbitrary axes. The action of the **rotation operator** on a function ψ is illustrated in Figure 6.4.

* **Problem 6.1** Consider the parity operator in three dimensions.

(a) Show that $\hat{\Pi} \psi(\mathbf{r}) = \psi'(\mathbf{r}) = \psi(-\mathbf{r})$ is equivalent to a mirror reflection followed by a rotation.

(b) Show that, for ψ expressed in polar coordinates, the action of the parity operator is

$$\hat{\Pi} \psi(r, \theta, \phi) = \psi(r, \pi - \theta, \phi + \pi).$$

(c) Show that for the hydrogenic orbitals,

$$\hat{\Pi} \psi_{n\ell m}(r, \theta, \phi) = (-1)^\ell \psi_{n\ell m}(r, \theta, \phi).$$

That is, $\psi_{n\ell m}$ is an eigenstate of the parity operator, with eigenvalue $(-1)^\ell$. *Note:* This result actually applies to the stationary states of any central potential $V(\mathbf{r}) = V(r)$. For a central potential, the eigenstates may be written in the separable form $R_{n\ell}(r) Y_\ell^m(\theta, \phi)$ where only the radial function $R_{n\ell}$ —which plays no role in determining the parity of the state—depends on the specific functional form of $V(r)$.

6.2 THE TRANSLATION OPERATOR

Equation 6.1 *defines* the translation operator. We can express $\hat{T}(a)$ in terms of the momentum operator, to which it is intimately related. To that end, we replace $\psi(x - a)$ by its Taylor series³

$$\begin{aligned}\hat{T}(a) \psi(x) &= \psi(x - a) = \sum_{n=0}^{\infty} \frac{1}{n!} (-a)^n \frac{d^n}{dx^n} \psi(x) \\ &= \sum_{n=0}^{\infty} \frac{1}{n!} \left(\frac{-i a}{\hbar} \hat{p} \right)^n \psi(x).\end{aligned}$$

The right-hand side of this equation is the exponential function,⁴ so

$$\hat{T}(a) = \exp \left[-\frac{i a}{\hbar} \hat{p} \right]. \quad (6.3)$$

We say that momentum is the “**generator**” of translations.⁵

Note that $\hat{T}(a)$ is a unitary operator:⁶

$$\hat{T}(a)^{-1} = \hat{T}(-a) = \hat{T}(a)^\dagger. \quad (6.4)$$

The first equality is obvious physically (the inverse operation of shifting something to the right is shifting it by an equal amount to the left), and the second equality then follows from taking the adjoint of Equation 6.3 (see Problem 6.2).

* **Problem 6.2** Show that, for a Hermitian operator $\hat{Q}$, the operator $\hat{U} = \exp [i \hat{Q}]$ is unitary.

Hint: First you need to prove that the adjoint is given by $\hat{U}^\dagger = \exp [-i \hat{Q}]$; then prove that $\hat{U}^\dagger \hat{U} = 1$. Problem 3.5 may help.

6.2.1 How Operators Transform

So far I have shown how to translate a *function*; this has an obvious graphical interpretation via Figure 6.2. We can also consider what it means to *translate an operator*. The transformed operator $\hat{Q}'$ is defined to be the operator that gives the same expectation value in the untranslated state ψ as does the operator $\hat{Q}$ in the translated state ψ' :

$$\langle \psi' | \hat{Q} | \psi' \rangle = \langle \psi | \hat{Q}' | \psi \rangle.$$

³ I'm assuming that our function has a Taylor series expansion, but the final result applies more generally. See Problem 6.31 for the details.

⁴ See Section 3.6.2 for the definition of the exponential of an operator.

⁵ The term comes from the study of **Lie groups** (the group of translations is an example). If you're interested, an introduction to Lie groups (written for physicists) can be found in George B. Arfken, Hans J. Weber, and Frank E. Harris, *Mathematical Methods for Physicists*, 7th edn, Academic Press, New York (2013), Section 17.7.

⁶ Unitary operators are discussed in Problem A.30. A unitary operator is one whose adjoint is also its inverse: $\hat{U}^\dagger \hat{U} = \hat{U} \hat{U}^\dagger = 1$.

There are two ways to calculate the effect of a translation on an expectation value. One could actually shift the wave function over some distance (this is called an **active transformation**) *or* one could leave the wave function where it was and shift the origin of our coordinate system by the same amount in the opposite direction (a **passive transformation**). The operator $\hat{Q}'$ is the operator in this shifted coordinate system.

Using Equation 6.1,

$$\langle \psi | \hat{T}^\dagger \hat{Q} \hat{T} | \psi \rangle = \langle \psi | \hat{Q}' | \psi \rangle. \quad (6.5)$$

Here I am using the fact that the adjoint of an operator is defined such that, if $\hat{T} | f \rangle \equiv | T f \rangle$, then $\langle T f | = \langle f | \hat{T}^\dagger$ (see Problem 3.5). Because Equation 6.5 is to hold for *all* ψ , it follows that

$$\boxed{\hat{Q}' = \hat{T}^\dagger \hat{Q} \hat{T}.} \quad (6.6)$$

The transformed operator for the case $\hat{Q} = \hat{x}$ is worked out in Example 6.1. Figure 6.5 illustrates the equivalence of the two ways of carrying out the transformation.

Example 6.1

Find the operator $\hat{x}'$ obtained by applying a translation through a distance a to the operator $\hat{x}$. That is, what is the action of $\hat{x}'$, as defined by Equation 6.6, on an arbitrary $f(x)$?

Solution: Using the definition of $\hat{x}'$ (Equation 6.6) and a test function $f(x)$ we have

$$\hat{x}' f(x) = \hat{T}^\dagger(a) \hat{x} \hat{T}(a) f(x),$$

and since $\hat{T}^\dagger(a) = \hat{T}(-a)$ (Equation 6.4),

$$\hat{x}' f(x) = \hat{T}(-a) \hat{x} \hat{T}(a) f(x).$$

From Equation 6.1

$$\hat{x}' f(x) = \hat{T}(-a) x f(x - a),$$

and from Equation 6.1 again, $\hat{T}(-a) [x f(x - a)] = (x + a) f(x)$, so

$$\hat{x}' f(x) = (x + a) f(x).$$

Finally we may read off the operator

$$\hat{x}' = \hat{x} + a. \quad (6.7)$$

As expected, Equation 6.7 corresponds to shifting the origin of our coordinates to the left by a so that positions in these transformed coordinates are greater by a than in the untransformed coordinates.

In Problem 6.3 you will apply a translation to the momentum operator to show that $\hat{p}' = \hat{T}^\dagger \hat{p} \hat{T} = \hat{p}$: the momentum operator is unchanged by this transformation. Physically, this is because the particle's momentum is independent of where you place the origin of your coordinates, depending only on *differences* in position: $p = m dx/dt$. Once you know how the position and momentum operators behave under a translation, you know how *any* operator does, since

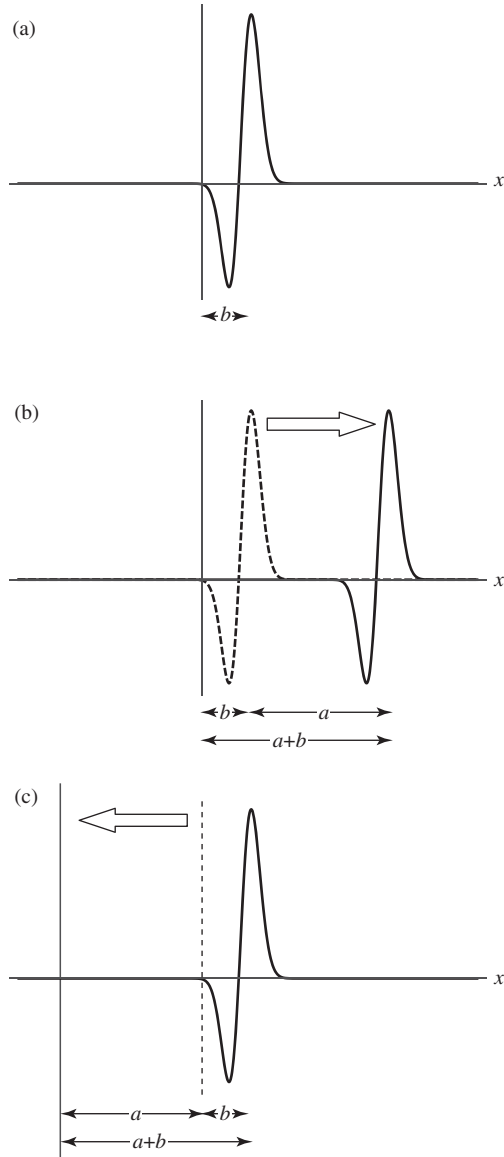


Figure 6.5: Active vs. passive transformations: (a) depicts the original function, (b) illustrates an active transformation in which the function is shifted to the right by an amount a , and (c) illustrates a passive transformation where the axes are shifted to the left by an amount a . A point on the wave a distance b from the origin before the transformation is a distance $a + b$ from the origin after the transformation in either (b) or (c); this is the equivalence of the two pictures.

$$\hat{Q}'(\hat{x}, \hat{p}) = \hat{T}^\dagger \hat{Q}(\hat{x}, \hat{p}) \hat{T} = \hat{Q}(\hat{x}', \hat{p}') = \hat{Q}(\hat{x} + a, \hat{p}). \quad (6.8)$$

Problem 6.4 will walk you through the proof.

Problem 6.3 Show that the operator $\hat{p}'$ obtained by applying a translation to the operator $\hat{p}$ is $\hat{p}' = \hat{T}^\dagger \hat{p} \hat{T} = \hat{p}$.

Problem 6.4 Prove Equation 6.8. You may assume that $Q(\hat{x}, \hat{p})$ can be written in a power series

$$\hat{Q}(\hat{x}, \hat{p}) = \sum_{m=0}^{\infty} \sum_{n=0}^{\infty} a_{mn} \hat{x}^m \hat{p}^n$$

for some constants a_{mn} .

6.2.2 Translational Symmetry

So far we have seen how a *function* behaves under a translation and how an *operator* behaves under a translation. I am now in a position to make precise the notion of a symmetry that I mentioned in the introduction. A system is **translationally invariant** (equivalent to saying it has translational symmetry) if the Hamiltonian is unchanged by the transformation:

$$\hat{H}' = \hat{T}^\dagger \hat{H} \hat{T} = \hat{H}.$$

Because $\hat{T}$ is unitary (Equation 6.4) we can multiply both sides of this equation by $\hat{T}$ to get

$$\hat{H} \hat{T} = \hat{T} \hat{H}.$$

Therefore, a system has translational symmetry if the Hamiltonian commutes with the translation operator:

$$[\hat{H}, \hat{T}] = 0. \quad (6.9)$$

For a particle of mass m moving in a one-dimensional potential, the Hamiltonian is

$$\hat{H} = \frac{\hat{p}^2}{2m} + V(x).$$

According to Equation 6.8, the transformed Hamiltonian is

$$\hat{H}' = \frac{\hat{p}^2}{2m} + V(x + a)$$

so translational symmetry implies that

$$V(x + a) = V(x). \quad (6.10)$$

Now, there are two very different physical settings where Equation 6.10 might arise. The first is a constant potential, where Equation 6.10 holds for *every* value of a ; such a system is said to have **continuous translational symmetry**. The second is a periodic potential, such as an electron might encounter in a crystal, where Equation 6.10 holds only for a discrete set of a s; such a system is said to have **discrete translational symmetry**. The two cases are illustrated in Figure 6.6.

DISCRETE TRANSLATIONAL SYMMETRY AND BLOCH'S THEOREM

What are the implications of translational symmetry? For a system with a discrete translational symmetry, the most important consequence is Bloch's theorem; the theorem specifies the form taken by the stationary states. We used this theorem in Section 5.3.2; I will now prove it.

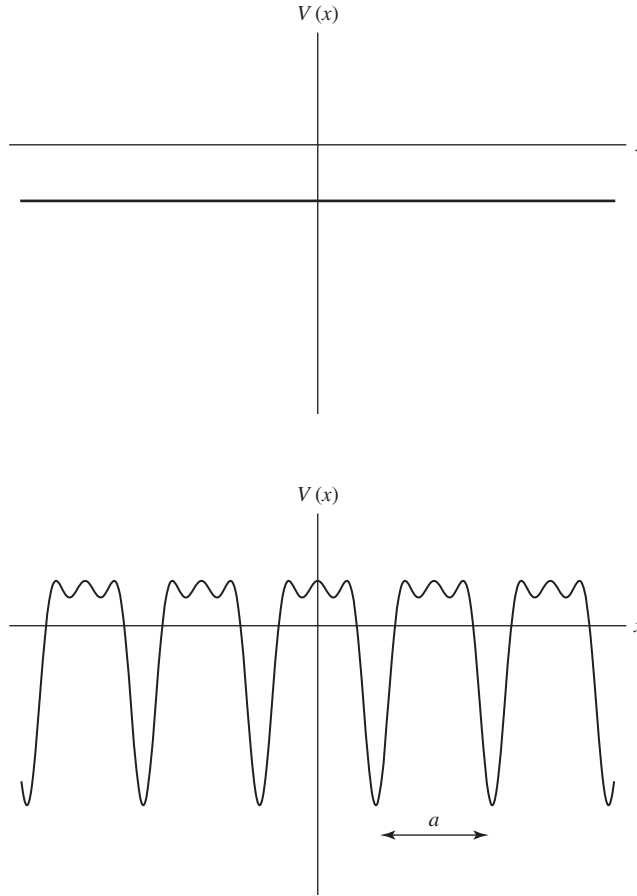


Figure 6.6: Potentials for a system with continuous (top) and discrete (bottom) translational symmetry. In the former case the potential is the same when shifted right or left by *any* amount; in the latter case the potential is the same when shifted right or left by an integer multiple of a .

In Section A.5 it is shown that if two operators commute, then they have a complete set of simultaneous eigenstates. This means that if the Hamiltonian is translationally invariant (which is to say, if it commutes with the translation operator), then the eigenstates $\psi(x)$ of the Hamiltonian can be chosen to be simultaneously eigenstates of $\hat{T}$:

$$\hat{H} \psi(x) = E \psi(x), \quad \hat{T}(a) \psi(x) = \lambda \psi(x),$$

where λ is the eigenvalue associated with $\hat{T}$. Since $\hat{T}$ is unitary, its eigenvalues have magnitude 1 (see Problem A.30), which means that λ can be written as $\lambda = e^{i\phi}$ for some real number ϕ . By convention we write $\phi = -qa$ where $\hbar q$ is called the **crystal momentum**. Therefore, the stationary states of a particle of mass m moving in a periodic potential have the property

$$\psi(x - a) = e^{-iqa} \psi(x). \quad (6.11)$$

There is a more illuminating way to write Equation 6.11:⁷

$$\boxed{\psi(x) = e^{i q x} u(x)} \quad (6.12)$$

where $u(x)$ is a periodic function of x : $u(x + a) = u(x)$ and $e^{i q x}$ is a traveling wave (recall that a traveling wave by itself describes a *free* particle—Section 2.4) with wavelength $2\pi/q$. Equation 6.12 is **Bloch's theorem** and it says that the stationary states of a particle in a periodic potential are periodic functions multiplying traveling waves. Note that just because the Hamiltonian is translationally invariant, that doesn't mean the stationary states *themselves* are translationally invariant, it simply means that they can be chosen to be eigenstates of the translation operator.

Bloch's theorem is truly remarkable. It tells us that the stationary states of a particle in a periodic potential (such as an electron in a crystal) are, apart from a periodic modulation, traveling waves. As such, they have a nonzero velocity.⁸ This means that an electron could travel through a perfect crystal without scattering! That has dramatic implications for electronic conduction in solids.

CONTINUOUS TRANSLATIONAL SYMMETRY AND MOMENTUM CONSERVATION

If a system has continuous translation symmetry then the Hamiltonian commutes with $\hat{T}(a)$ for any choice of a . In this case it is useful to consider an **infinitesimal translation**

$$\hat{T}(\delta) = e^{-i \delta \hat{p}/\hbar} \approx 1 - i \frac{\delta}{\hbar} \hat{p},$$

where δ is an infinitesimal length.⁹

If the Hamiltonian has continuous translational symmetry, then it must be unchanged under any translation, including an infinitesimal one; equivalently it commutes with the translation operator, and hence

$$[\hat{H}, \hat{T}(\delta)] = \left[\hat{H}, 1 - i \frac{\delta}{\hbar} \hat{p} \right] = 0 \Rightarrow [\hat{H}, \hat{p}] = 0.$$

So if the Hamiltonian has continuous translational symmetry, it must commute with the momentum operator. And if the Hamiltonian commutes with momentum, then according to the “generalized Ehrenfest's theorem” (Equation 3.73)

$$\frac{d}{dt} \langle p \rangle = \frac{i}{\hbar} \left\langle [\hat{H}, \hat{p}] \right\rangle = 0. \quad (6.13)$$

⁷ It is clear that Equation 6.12 satisfies Equation 6.11. In Problem 6.5 you'll prove that they are in fact equivalent statements.

⁸ For a delightful proof using perturbation theory, see Neil Ashcroft and N. David Mermin, *Solid State Physics*, Cengage, Belmont, 1976 (p. 765), after you have completed Problem 6.6 and studied Chapter 7.

⁹ For the case of continuous symmetries, it is often much easier to work with the infinitesimal form of the transformation; any finite transformation can then be built up as a product of infinitesimal transformations. In particular, the finite translation by a is a sequence of N infinitesimal translations with $\delta = a/N$ in the limit that $N \rightarrow \infty$:

$$\lim_{N \rightarrow \infty} \left(1 - i \frac{a}{N} \frac{1}{\hbar} \hat{p} \right)^N = \exp \left[-i \frac{a}{\hbar} \hat{p} \right].$$

This is a statement of **momentum conservation** and we have now shown that continuous translational symmetry implies that momentum is conserved. This is our first example of a powerful general principle: *symmetries imply conservation laws*.¹⁰

Of course, if we're talking about a single particle of mass m moving in a potential $V(x)$, the only potential that has continuous translational symmetry is the constant potential, which is equivalent to the free particle. And it is pretty obvious that momentum is conserved in that case. But the analysis here readily extends to a system of interacting particles (see Problem 6.7). The fact that momentum is conserved in that case as well (so long as the Hamiltonian is translationally invariant) is a highly nontrivial result. In any event, the point to remember is that *conservation of momentum is a consequence of translational symmetry*.

Problem 6.5 Show that Equation 6.12 follows from Equation 6.11. *Hint:* First write $\psi(x) = e^{iqx} u(x)$, which is certainly true for *some* $u(x)$, and then show that $u(x)$ is necessarily a periodic function of x .



Problem 6.6 Consider a particle of mass m moving in a potential $V(x)$ with period a . We know from Bloch's theorem that the wave function can be written in the form of Equation 6.12. *Note:* It is conventional to label the states with quantum numbers n and q as $\psi_{nq}(x) = e^{-iqx} u_{nq}(x)$ where E_{nq} is the n th energy for a given value of q .

(a) Show that u satisfies the equation

$$-\frac{\hbar^2}{2m} \frac{d^2 u_{nq}}{dx^2} - \frac{i\hbar^2 q}{m} \frac{du_{nq}}{dx} + V(x) u_{nq} = \left(E_{nq} - \frac{\hbar^2 q^2}{2m} \right) u_{nq}. \quad (6.14)$$

(b) Use the technique from Problem 2.61 to solve the differential equation for u_{nq} . You need to use a two-sided difference for the first derivative so that you have a Hermitian matrix to diagonalize: $\frac{d\psi}{dx} \approx \frac{\psi_{j+1} - \psi_{j-1}}{2\Delta x}$. For the potential in the interval 0 to a let

$$V(x) = \begin{cases} -V_0 & a/4 < x < 3a/4 \\ 0 & \text{else} \end{cases}$$

with $V_0 = 20 \hbar^2 / 2m a^2$. (You will need to modify the technique slightly to account for the fact that the function u_{nq} is periodic.) Find the lowest two energies for the following values of the crystal momentum: $qa = -\pi, -\pi/2, 0, \pi/2, \pi$. Note that q and $q + 2\pi/a$ describe the same wave function (Equation 6.12), so there is no reason to consider values of qa outside of the interval from $-\pi$ to π . In solid state physics, the values of q inside this range constitute the **first Brillouin zone**.

(c) Make a plot of the energies E_{1q} and E_{2q} for values of q between $-\pi/a$ and π/a . If you've automated the code that you used in part (b), you should be able to show a large number of q values in this range. If not, simply plot the values that you computed in (b).

¹⁰ In the case of a discrete translational symmetry, momentum is not conserved, but there is a conserved quantity closely related to the discrete translational symmetry, which is the crystal momentum. For a discussion of crystal momentum see Steven H. Simon, *The Oxford Solid State Basics*, Oxford, 2013, p.84.

Problem 6.7 Consider two particles of mass m_1 and m_2 (in one dimension) that interact via a potential that depends only on the distance between the particles $V(|x_1 - x_2|)$, so that the Hamiltonian is

$$\hat{H} = -\frac{\hbar^2}{2m_1} \frac{\partial^2}{\partial x_1^2} - \frac{\hbar^2}{2m_2} \frac{\partial^2}{\partial x_2^2} + V(|x_1 - x_2|).$$

Acting on a two-particle wave function the translation operator would be

$$\hat{T}(a) \psi(x_1, x_2) = \psi(x_1 - a, x_2 - a).$$

(a) Show that the translation operator can be written

$$\hat{T}(a) = e^{-\frac{ia}{\hbar} \hat{P}},$$

where $\hat{P} = \hat{p}_1 + \hat{p}_2$ is the total momentum.

(b) Show that the total momentum is conserved for this system.

6.3 CONSERVATION LAWS

In classical mechanics the meaning of a conservation law is straightforward: the quantity in question is the same before and after some event. Drop a rock, and potential energy is converted into kinetic energy, but the *total* is the same just before it hits the ground as when it was released; collide two billiard balls and momentum is transferred from one to the other, but the total remains unchanged. But in quantum mechanics a system does not in general *have* a definite energy (or momentum) before the process begins (or afterward). What does it *mean*, in that case, to say that the observable Q is (or is not) conserved? Here are two possibilities:

- **First definition:** The *expectation value* $\langle Q \rangle$ is independent of time.
- **Second definition:** The probability of getting any particular value is independent of time.

Under what conditions does each of these conservation laws hold?

Let us stipulate that the observable in question does not depend explicitly on time: $\partial Q / \partial t = 0$. In that case the generalized Ehrenfest theorem (Equation 3.73) tells us that the expectation value of Q is independent of time if *The operator $\hat{Q}$ commutes with the Hamiltonian*. It so happens that the same criterion guarantees conservation by the second definition.

I will now prove this result. Recall that the probability of getting the result q_n in a measurement of Q at time t is (Equation 3.43)

$$P(q_n) = |\langle f_n | \Psi(t) \rangle|^2, \quad (6.15)$$

where f_n is the corresponding eigenvector: $\hat{Q} |f_n\rangle = q_n |f_n\rangle$.¹¹ We know that the time evolution of the wave function is (Equation 2.17)

¹¹ If the spectrum of $\hat{Q}$ is degenerate (there are distinct eigenvectors with the same eigenvalue q_n : $\hat{Q} |f_n^{(i)}\rangle = q_n |f_n^{(i)}\rangle$ for $i = 1, 2, \dots$), then we need to sum over those states:

$$P(q_n) = \sum_i \left| \langle f_n^{(i)} | \Psi(t) \rangle \right|^2.$$

Except for the sum over i the proof proceeds unchanged.

$$|\Psi(t)\rangle = \sum_m e^{-i E_m t/\hbar} c_m |\psi_m\rangle,$$

where the $|\psi_n\rangle$ are the eigenstates of $\hat{H}$, and therefore

$$P(q_n) = \left| \sum_m e^{-i E_m t/\hbar} c_m \langle f_n | \psi_m \rangle \right|^2.$$

Now the key point: since $\hat{Q}$ and $\hat{H}$ commute we can find a complete set of simultaneous eigenstates for them (see Section A.5); without loss of generality then $|f_n\rangle = |\psi_n\rangle$. Using the orthonormality of the $|\psi_n\rangle$,

$$P(q_n) = \left| \sum_m e^{-i E_m t/\hbar} c_m \langle \psi_n | \psi_m \rangle \right|^2 = |c_n|^2,$$

which is clearly independent of time.

6.4 PARITY

6.4.1 Parity in One Dimension

A spatial inversion is implemented by the parity operator $\hat{\Pi}$; in one dimension,

$$\hat{\Pi} \psi(x) = \psi'(x) = \psi(-x).$$

Evidently, the parity operator is its own inverse: $\hat{\Pi}^{-1} = \hat{\Pi}$; in Problem 6.8 you will show that it is Hermitian: $\hat{\Pi}^\dagger = \hat{\Pi}$. Putting this together, the parity operator is unitary as well:

$$\hat{\Pi}^{-1} = \hat{\Pi} = \hat{\Pi}^\dagger. \quad (6.16)$$

Operators transform under a spatial inversion as

$$\hat{Q}' = \hat{\Pi}^\dagger \hat{Q} \hat{\Pi}. \quad (6.17)$$

I won't repeat the argument leading up to Equation 6.17, since it is identical to the one by which we arrived at Equation 6.6 in the case of translations. The position and momentum operators are “odd under parity” (Problem 6.10):

$$\hat{x}' = \hat{\Pi}^\dagger \hat{x} \hat{\Pi} = -\hat{x}, \quad (6.18)$$

$$\hat{p}' = \hat{\Pi}^\dagger \hat{p} \hat{\Pi} = -\hat{p}, \quad (6.19)$$

and this tells us how *any* operator transforms (see Problem 6.4):

$$\hat{Q}'(\hat{x}, \hat{p}) = \hat{\Pi}^\dagger \hat{Q}(\hat{x}, \hat{p}) \hat{\Pi} = \hat{Q}(-\hat{x}, -\hat{p}).$$

A system has **inversion symmetry** if the Hamiltonian is unchanged by a parity transformation:

$$\hat{H}' = \hat{\Pi}^\dagger \hat{H} \hat{\Pi} = \hat{H},$$

or, using the unitarity of the parity operator,

$$[\hat{H}, \hat{\Pi}] = 0. \quad (6.20)$$

If our Hamiltonian describes a particle of mass m in a one-dimensional potential $V(x)$, then inversion symmetry simply means that the potential is an even function of position:

$$V(x) = V(-x).$$

The implications of inversion symmetry are two: First, we can find a complete set of simultaneous eigenstates of $\hat{\Pi}$ and $\hat{H}$. Let such an eigenstate be written ψ_n ; it satisfies

$$\hat{\Pi} \psi_n(x) = \psi_n(-x) = \pm \psi_n(x),$$

since the eigenvalues of the parity operator are restricted to ± 1 (Problem 6.8). So the stationary states of a potential that is an even function of position are themselves even or odd functions (or can be chosen as such, in the case of degeneracy).¹² This property is familiar from the simple harmonic oscillator, the infinite square well (if the origin is placed at the center of the well), and the Dirac delta function potential, and you proved it in general in Problem 2.1.

Second, according to Ehrenfest's theorem, if the Hamiltonian has an inversion symmetry then

$$\frac{d}{dt} \langle \Pi \rangle = \frac{i}{\hbar} \langle [\hat{H}, \hat{\Pi}] \rangle = 0$$

so *parity is conserved* for a particle moving in a symmetric potential. And not just the expectation value, but the *probability* of any particular outcome in a measurement, in accord with the theorem of Section 6.3. Parity conservation means, for example, that if the wave function of a particle in a harmonic oscillator potential is even at $t = 0$ then it will be even at any later time t ; see Figure 6.7.

*

Problem 6.8

- (a) Show that the parity operator $\hat{\Pi}$ is Hermitian.
 (b) Show that the eigenvalues of the parity operator are ± 1 .

6.4.2 Parity in Three Dimensions

The spatial inversion generated by the parity operator in three dimensions is

$$\hat{\Pi} \psi(\mathbf{r}) = \psi'(\mathbf{r}) = \psi(-\mathbf{r}).$$

The operators $\hat{\mathbf{r}}$ and $\hat{\mathbf{p}}$ transform as

$$\hat{\mathbf{r}}' = \hat{\Pi}^\dagger \hat{\mathbf{r}} \hat{\Pi} = -\hat{\mathbf{r}}, \quad (6.21)$$

$$\hat{\mathbf{p}}' = \hat{\Pi}^\dagger \hat{\mathbf{p}} \hat{\Pi} = -\hat{\mathbf{p}}. \quad (6.22)$$

Any other operator transforms as

$$\hat{Q}'(\hat{\mathbf{r}}, \hat{\mathbf{p}}) = \hat{\Pi}^\dagger \hat{Q}(\hat{\mathbf{r}}, \hat{\mathbf{p}}) \hat{\Pi} = \hat{Q}(-\hat{\mathbf{r}}, -\hat{\mathbf{p}}). \quad (6.23)$$

¹² For bound (normalizable) states in one dimension, there is no degeneracy and every bound state of a symmetric potential is automatically an eigenstate of parity. (However, see Problem 2.46.) For scattering states, degeneracy does occur.

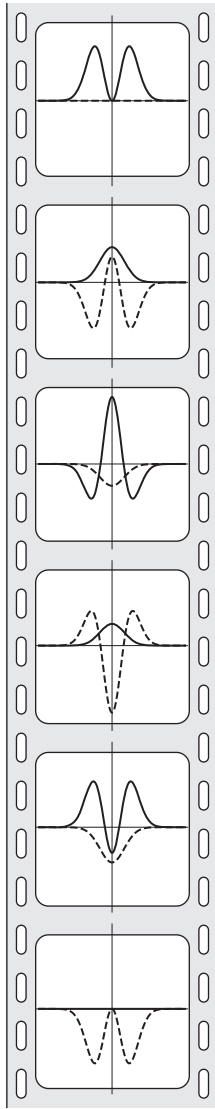


Figure 6.7: This filmstrip shows the time evolution of a particular wave function ($\Psi_0(\xi) = A \xi^2 e^{-\xi^2/2}$) for a particle in the harmonic oscillator potential. The solid and dashed curves are the real and imaginary parts of the wave function respectively, and time increases from top to bottom. Since parity is conserved, a wave function which is initially an even function of position (as this one is) remains an even function at all later times.

Example 6.2

Find the parity-transformed angular momentum operator $\hat{\mathbf{L}}' = \hat{\Pi}^\dagger \hat{\mathbf{L}} \hat{\Pi}$, in terms of $\hat{\mathbf{L}}$.

Solution: Since $\mathbf{L} = \mathbf{r} \times \mathbf{p}$, Equation 6.23 tells us that

$$\hat{\mathbf{L}}' = \hat{\Pi}^\dagger \hat{\mathbf{L}} \hat{\Pi} = \hat{\mathbf{r}}' \times \hat{\mathbf{p}}' = (-\hat{\mathbf{r}}) \times (-\hat{\mathbf{p}}) = \hat{\mathbf{r}} \times \hat{\mathbf{p}} = \hat{\mathbf{L}}. \quad (6.24)$$

We have a special name for vectors like $\hat{\mathbf{L}}$, that are *even* under parity. We call them **pseudovectors**, since they don't change sign under parity the way “true” vectors, such as $\hat{\mathbf{r}}$ or $\hat{\mathbf{p}}$,

do. Similarly, scalars that are *odd* under parity are called **pseudoscalars**, since they do not behave under parity the way that “true” scalars (such as $\hat{\mathbf{r}} \cdot \hat{\mathbf{r}}$ which is even under parity) do. See Problem 6.9. *Note:* The labels **scalar** and **vector** describe how the operators behave under *rotations*; we will define these terms carefully in the next section. “True” vectors and pseudovectors behave the same way under a rotation—they are both vectors.

In three dimensions, the Hamiltonian for a particle of mass m moving in a potential $V(\mathbf{r})$ will have inversion symmetry if $V(-\mathbf{r}) = V(\mathbf{r})$. Importantly, any central potential satisfies this condition. As in the one-dimensional case, parity is conserved for such systems, and the eigenstates of the Hamiltonian may be chosen to be simultaneously eigenstates of parity. In Problem 6.1 you proved that the eigenstates of a particle in a central potential, written $\psi_{n\ell m}(r, \theta, \phi) = R_{n\ell}(r) Y_{\ell}^m(\theta, \phi)$, are eigenstates of parity:¹³

$$\hat{\Pi} \psi_{n\ell m}(r, \theta, \phi) = (-1)^{\ell} \psi_{n\ell m}(r, \theta, \phi).$$

*

Problem 6.9

(a) Under parity, a “true” scalar operator does not change:

$$\hat{\Pi}^{\dagger} \hat{f} \hat{\Pi} = \hat{f},$$

whereas a pseudoscalar changes sign. Show therefore that $[\hat{\Pi}, \hat{f}] = 0$ for a “true” scalar, whereas $\{\hat{\Pi}, \hat{f}\} = 0$ for a pseudoscalar. *Note:* the **anti-commutator** of two operators $\hat{A}$ and $\hat{B}$ is defined as $\{\hat{A}, \hat{B}\} \equiv \hat{A} \hat{B} + \hat{B} \hat{A}$.

(b) Similarly, a “true” vector changes sign

$$\hat{\Pi}^{\dagger} \hat{\mathbf{V}} \hat{\Pi} = -\hat{\mathbf{V}},$$

whereas a pseudovector is unchanged. Show therefore that $\{\hat{\Pi}, \hat{\mathbf{V}}\} = \mathbf{0}$ for a “true” vector and $[\hat{\Pi}, \hat{\mathbf{V}}] = \mathbf{0}$ for a pseudovector.

6.4.3 Parity Selection Rules

Selection rules tell you when a matrix element is zero based on the symmetry of the situation. Recall that a matrix element is any object of the form $\langle b | \hat{Q} | a \rangle$; an expectation value is a special case of a matrix element with $a = b = \psi$. One operator whose selection rules are physically important is the electric dipole moment operator

$$\hat{\mathbf{p}}_e = q \hat{\mathbf{r}}.$$

The selection rules for this operator—the operator itself is nothing more than the charge of the particle times its position—determine which atomic transitions are allowed and which are forbidden (see Chapter 11). It is odd under parity since the position vector $\hat{\mathbf{r}}$ is odd:

¹³ Note that Equation 6.24 could equivalently be written as $[\hat{\Pi}, \hat{\mathbf{L}}] = \mathbf{0}$. The fact that parity commutes with every component of the angular momentum (and therefore also $\hat{L}^2$) is the reason you can find simultaneous eigenstates of $\hat{L}^2$, $\hat{L}_z$, and $\hat{\Pi}$.

$$\hat{\Pi}^\dagger \hat{\mathbf{p}}_e \hat{\Pi} = -\hat{\mathbf{p}}_e. \quad (6.25)$$

Now consider the matrix elements of the electric dipole operator between two states $\psi_{n\ell m}$ and $\psi_{n'\ell'm'}$ (we label the corresponding kets $|n\ell m\rangle$ and $|n'\ell'm'\rangle$). Using Equation 6.25 we have

$$\begin{aligned} \langle n'\ell'm' | \hat{\mathbf{p}}_e | n\ell m \rangle &= -\langle n'\ell'm' | \hat{\Pi}^\dagger \hat{\mathbf{p}}_e \hat{\Pi} | n\ell m \rangle \\ &= -\langle n'\ell'm' | (-1)^{\ell'} \hat{\mathbf{p}}_e (-1)^\ell | n\ell m \rangle \\ &= (-1)^{\ell+\ell'+1} \langle n'\ell'm' | \hat{\mathbf{p}}_e | n\ell m \rangle. \end{aligned}$$

From this we see immediately that

$$\langle n'\ell'm' | \hat{\mathbf{p}}_e | n\ell m \rangle = 0 \text{ if } \ell + \ell' \text{ is even.} \quad (6.26)$$

This is called **Laporte's rule**; it says that matrix elements of the dipole moment operator vanish between states with the same parity. The reasoning by which we obtained Equation 6.26 can be generalized to derive selection rules for any operator, as long as you know how that operator transforms under parity. In particular, *Laporte's rule applies to any operator that is odd under parity*. The selection rule for an operator that is *even* under parity, such as $\hat{\mathbf{L}}$, is derived in Problem 6.11.

Problem 6.10 Show that the position and momentum operators are odd under parity. That is, prove Equations 6.18, 6.19, and, by extension, 6.21 and 6.22.

*

Problem 6.11 Consider the matrix elements of $\hat{\mathbf{L}}$ between two definite-parity states: $\langle n'\ell'm' | \hat{\mathbf{L}} | n\ell m \rangle$. Under what conditions is this matrix element guaranteed to vanish? Note that the same selection rule would apply to any pseudovector operator, or any “true” scalar operator.

Problem 6.12 Spin angular momentum, $\hat{\mathbf{S}}$, is even under parity, just like orbital angular momentum $\hat{\mathbf{L}}$:

$$\hat{\Pi}^\dagger \hat{\mathbf{S}} \hat{\Pi} = \hat{\mathbf{S}} \quad \text{or} \quad [\hat{\Pi}, \hat{\mathbf{S}}] = 0. \quad (6.27)$$

Acting on a spinor written in the standard basis (Equation 4.139), the parity operator becomes a 2×2 matrix. Show that, due to Equation 6.27, this matrix must be a constant times the identity matrix. As such, the parity of a spinor isn't very interesting since both spin states are parity eigenstates with the same eigenvalue. We can arbitrarily choose that parity to be $+1$, so the parity operator has no effect on the spin portion of the wave function.¹⁴

¹⁴ However, it turns out that *antiparticles* of spin $1/2$ have opposite parity. Thus the electron is conventionally assigned parity $+1$, but the positron then has parity -1 .

* **Problem 6.13** Consider an electron in a hydrogen atom.

(a) Show that if the electron is in the ground state, then necessarily $\langle \mathbf{p}_e \rangle = 0$. No calculation allowed.

(b) Show that if the electron is in an $n = 2$ state, then $\langle \mathbf{p}_e \rangle$ need not vanish. Give an example of a wave function for the energy level $n = 2$ that has a non-vanishing $\langle \mathbf{p}_e \rangle$ and compute $\langle \mathbf{p}_e \rangle$ for this state.

6.5 ROTATIONAL SYMMETRY

6.5.1 Rotations About the z Axis

The operator that rotates a function about the z axis by an angle φ (Equation 6.2)

$$\hat{R}_z(\varphi) \psi(r, \theta, \phi) = \psi'(r, \theta, \phi) = \psi(r, \theta, \phi - \varphi) \quad (6.28)$$

is closely related to the z component of angular momentum (Equation 4.129). By the same reasoning that led to Equation 6.3,

$$\hat{R}_z(\varphi) = \exp \left[-\frac{i\varphi}{\hbar} \hat{L}_z \right], \quad (6.29)$$

and we say that $\hat{L}_z$ is the **generator of rotations** about the z axis (compare Equation 6.3).

How do the operators $\hat{\mathbf{r}}$ and $\hat{\mathbf{p}}$ transform under rotations? To answer this question we use the infinitesimal form of the operator:

$$\hat{R}_z(\delta) \approx 1 - \frac{i\delta}{\hbar} \hat{L}_z.$$

Then the operator $\hat{x}$ transforms as

$$\begin{aligned} \hat{x}' &= \hat{R}^\dagger \hat{x} \hat{R} \approx \left(1 + \frac{i\delta}{\hbar} \hat{L}_z \right) \hat{x} \left(1 - \frac{i\delta}{\hbar} \hat{L}_z \right) \\ &\approx \hat{x} + \frac{i\delta}{\hbar} [\hat{L}_z, \hat{x}] \approx \hat{x} - \delta \hat{y} \end{aligned}$$

(I used Equation 4.122 for the commutator). Similar calculations show that $\hat{y}' = \hat{y} + \delta \hat{x}$ and $\hat{z}' = \hat{z}$. We can combine these results into a matrix equation

$$\begin{pmatrix} \hat{x}' \\ \hat{y}' \\ \hat{z}' \end{pmatrix} = \begin{pmatrix} 1 & -\delta & 0 \\ \delta & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} \hat{x} \\ \hat{y} \\ \hat{z} \end{pmatrix}. \quad (6.30)$$

That doesn't look quite right for a rotation. Shouldn't it be

$$\begin{pmatrix} \hat{x}' \\ \hat{y}' \\ \hat{z}' \end{pmatrix} = \begin{pmatrix} \cos \varphi & -\sin \varphi & 0 \\ \sin \varphi & \cos \varphi & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} \hat{x} \\ \hat{y} \\ \hat{z} \end{pmatrix} ? \quad (6.31)$$

Yes, but don't forget, we are assuming $\varphi \rightarrow \delta$ is infinitesimal, so (dropping terms of order δ^2 and higher) $\cos \varphi \rightarrow 1$ and $\sin \varphi \rightarrow \delta$.¹⁵

¹⁵ To go the other way, from infinitesimal to finite, see Problem 6.14.

**

Problem 6.14 In this problem you will establish the correspondence between Equations 6.30 and 6.31.

(a) Diagonalize the matrix¹⁶

$$\mathbf{M} = \begin{pmatrix} 1 & -\varphi/N \\ \varphi/N & 1 \end{pmatrix}$$

to obtain the matrix

$$\mathbf{M}' = \mathbf{S} \mathbf{M} \mathbf{S}^{-1},$$

where $\mathbf{S}^{-1}$ is the unitary matrix whose columns are the (normalized) eigenvectors of $\mathbf{M}$.

(b) Use the binomial expansion to show that $\lim_{N \rightarrow \infty} (\mathbf{M}')^N$ is a diagonal matrix with entries $e^{-i\varphi}$ and $e^{i\varphi}$ on the diagonal.

(c) Transform back to the original basis to show that

$$\lim_{N \rightarrow \infty} \mathbf{M}^N = \mathbf{S}^{-1} \left[\lim_{N \rightarrow \infty} (\mathbf{M}')^N \right] \mathbf{S}$$

agrees with the matrix in Equation 6.31.

6.5.2 Rotations in Three Dimensions

Equation 6.29 can be generalized in the obvious way to a rotation about an axis along the unit vector $\mathbf{n}$:

$$\hat{R}_{\mathbf{n}}(\varphi) = \exp \left[-\frac{i\varphi}{\hbar} \mathbf{n} \cdot \hat{\mathbf{L}} \right] \quad (6.32)$$

Just as linear momentum is the generator of translations, angular momentum is the **generator of rotations**.

Any operator (with three components) that transforms the same way as the position operator under rotations is called a **vector operator**. By “transforms the same way” we mean that $\hat{\mathbf{V}}' = \mathbf{D} \hat{\mathbf{V}}$ where $\mathbf{D}$ is the same matrix as appears in $\hat{\mathbf{r}}' = \mathbf{D} \hat{\mathbf{r}}$. In particular for a rotation about the z axis, we would have (Equation 6.31)

$$\begin{pmatrix} \hat{V}'_x \\ \hat{V}'_y \\ \hat{V}'_z \end{pmatrix} = \begin{pmatrix} \cos \varphi & -\sin \varphi & 0 \\ \sin \varphi & \cos \varphi & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} \hat{V}_x \\ \hat{V}_y \\ \hat{V}_z \end{pmatrix}.$$

This transformation rule follows from the commutation relations¹⁷

$$[\hat{L}_i, \hat{V}_j] = i \hbar \epsilon_{ijk} \hat{V}_k, \quad (6.33)$$

¹⁶ See Section A.5.

¹⁷ The Levi-Civita symbol ϵ_{ijk} is defined in Problem 4.29.

Table 6.1: Operators are classified as vectors or scalars based on their commutation relations with $\hat{\mathbf{L}}$, which encode how they transform under a rotation, and as pseudo- or “true” quantities based on their commutation relations with $\hat{\Pi}$, which encode how they transform under a spatial inversion. The curly brackets in the first column denote the anti-commutator, defined in Problem 6.9. To include the spin $\hat{\mathbf{S}}$ in this table, one simply replaces $\hat{L}_i$ everywhere it appears in the third column with $\hat{J}_i = \hat{L}_i + \hat{S}_i$ (Problems 6.12 and 6.32, respectively, discuss the effect of parity and rotations on spinors). $\hat{\mathbf{S}}$, like $\hat{\mathbf{L}}$, is then a pseudovector and $\hat{\mathbf{p}} \cdot \hat{\mathbf{S}}$ is a pseudoscalar.

	parity	rotations	examples
true vector $\hat{\mathbf{V}}$	$\{\hat{\Pi}, \hat{V}_i\} = 0$	$[\hat{L}_i, \hat{V}_j] = i \hbar \epsilon_{ijk} \hat{V}_k$	$\hat{\mathbf{r}}, \hat{\mathbf{p}}$
pseudovector $\hat{\mathbf{V}}$	$[\hat{\Pi}, \hat{V}_i] = 0$	$[\hat{L}_i, \hat{V}_j] = i \hbar \epsilon_{ijk} \hat{V}_k$	$\hat{\mathbf{L}}$
true scalar $\hat{f}$	$[\hat{\Pi}, \hat{f}] = 0$	$[\hat{L}_i, \hat{f}] = 0$	$\hat{\mathbf{r}} \cdot \hat{\mathbf{r}}$
pseudoscalar $\hat{f}$	$\{\hat{\Pi}, \hat{f}\} = 0$	$[\hat{L}_i, \hat{f}] = 0$	

(see Problem 6.16), and we may take Equation 6.33 as the *definition* of a vector operator. So far we have encountered three such operators, $\hat{\mathbf{r}}$, $\hat{\mathbf{p}}$ and $\hat{\mathbf{L}}$:

$$[\hat{L}_i, \hat{r}_j] = i \hbar \epsilon_{ijk} \hat{r}_k, \quad [\hat{L}_i, \hat{p}_j] = i \hbar \epsilon_{ijk} \hat{p}_k, \quad [\hat{L}_i, \hat{L}_j] = i \hbar \epsilon_{ijk} \hat{L}_k$$

(see Equations 4.99 and 4.122).

A **scalar operator** is a single quantity that is unchanged by rotations; this is equivalent to saying that the operator commutes with $\hat{\mathbf{L}}$:

$$\boxed{[\hat{L}_i, \hat{f}] = 0.} \quad (6.34)$$

We can now classify operators as either scalars or vectors, based on their commutation relations with $\hat{\mathbf{L}}$ (how they transform under a rotation), and as “true” or pseudo-quantities, based on their commutators with $\hat{\Pi}$ (how they transform under parity). These results are summarized in Table 6.1.¹⁸

CONTINUOUS ROTATIONAL SYMMETRY

For a particle of mass m moving in a potential $V(\mathbf{r})$, the Hamiltonian

$$\hat{H} = \frac{\hat{p}^2}{2m} + V(\mathbf{r})$$

is rotationally invariant if $V(\mathbf{r}) = V(r)$ (the central potentials studied in Section 4.1.1). In this case the Hamiltonian commutes with a rotation by any angle about an arbitrary axis

¹⁸ Of course, not every operator will fit into one of these categories. Scalar and vector operators are simply the first two instances in a hierarchy of tensor operators. Next come second-rank tensors (the inertia tensor from classical mechanics or the quadrupole tensor from electrodynamics are examples), third-rank tensors, and so forth.

$$[\hat{H}, \hat{R}_{\mathbf{n}}(\varphi)] = 0. \quad (6.35)$$

In particular, Equation 6.35 must hold for an *infinitesimal* rotation

$$R_{\mathbf{n}}(\delta) \approx 1 - i \frac{\delta}{\hbar} \mathbf{n} \cdot \hat{\mathbf{L}},$$

which means that the Hamiltonian commutes with the three components of $\mathbf{L}$:

$$[\hat{H}, \hat{\mathbf{L}}] = \mathbf{0}. \quad (6.36)$$

What, then, are the consequences of rotational invariance?

From Equation 6.36 and Ehrenfest's theorem

$$\frac{d}{dt} \langle \mathbf{L} \rangle = \frac{i}{\hbar} \langle [\hat{H}, \hat{\mathbf{L}}] \rangle = \mathbf{0} \quad (6.37)$$

for a central potential. Thus, *angular momentum conservation is a consequence of rotational invariance*. And beyond the statement 6.37, angular momentum conservation means that the probability distributions (for each component of the angular momentum) are independent of time as well—see Section 6.3.

Since the Hamiltonian for a central potential commutes with all three components of angular momentum, it also commutes with $\hat{L}^2$. The operators $\hat{H}$, $\hat{L}_z$, and $\hat{L}^2$ form a **complete set of compatible observables** for the bound states of a central potential. Compatible means that they commute pairwise

$$\begin{aligned} [\hat{H}, \hat{L}^2] &= 0, \\ [\hat{H}, \hat{L}_z] &= 0, \\ [\hat{L}_z, \hat{L}^2] &= 0, \end{aligned} \quad (6.38)$$

so that the eigenstates of $\hat{H}$ can be chosen to be simultaneous eigenstates of $\hat{L}^2$ and $\hat{L}_z$.

$$\begin{aligned} \hat{H} \psi_{n\ell m} &= E_n \psi_{n\ell m}, \\ \hat{L}_z \psi_{n\ell m} &= m \hbar \psi_{n\ell m}, \\ \hat{L}^2 \psi_{n\ell m} &= \ell(\ell+1) \hbar^2 \psi_{n\ell m}. \end{aligned}$$

Saying they are complete means that the quantum numbers n , ℓ , and m uniquely specify a bound state of the Hamiltonian. This is familiar from our solution to the hydrogen atom, the infinite spherical well, and the three-dimensional harmonic oscillator, but it is true for *any* central potential.¹⁹

*

Problem 6.15 Show how Equation 6.34 guarantees that a scalar is unchanged by a rotation:
 $\hat{f}' = \hat{R}^\dagger \hat{f} \hat{R} = \hat{f}.$

¹⁹ This follows from the fact that the radial Schrödinger equation (Equation 4.35) has at most a single normalizable solution so that, once you have specified ℓ and m , the energy uniquely specifies the state. The principal quantum number n indexes those energy values that lead to normalizable solutions.

- * **Problem 6.16** Working from Equation 6.33, find how the vector operator $\hat{\mathbf{V}}$ transforms for an infinitesimal rotation by an angle δ about the y axis. That is, find the matrix $\mathbf{D}$ in

$$\hat{\mathbf{V}}' = \mathbf{D} \hat{\mathbf{V}}.$$

Problem 6.17 Consider the action of an infinitesimal rotation about the $\mathbf{n}$ axis of an angular momentum eigenstate $\psi_{n\ell m}$. Show that

$$\hat{R}_{\mathbf{n}}(\delta) \psi_{n\ell m} = \sum_{m'} D_{m'm} \psi_{n\ell m'},$$

and find the complex numbers $D_{m'm}$ (they will depend on δ , $\mathbf{n}$, and ℓ as well as m and m'). This result makes sense: a rotation doesn't change the magnitude of the angular momentum (specified by ℓ) but does change its projection along the z axis (specified by m).

6.6 DEGENERACY

Symmetry is the source of most²⁰ degeneracy in quantum mechanics. We have seen that a symmetry implies the existence of an operator $\hat{Q}$ that commutes with the Hamiltonian

$$[\hat{H}, \hat{Q}] = 0. \quad (6.39)$$

So why does symmetry lead to degeneracy in the energy spectrum? The basic idea is this: if we have a stationary state $|\psi_n\rangle$, then $|\psi'_n\rangle = \hat{Q} |\psi_n\rangle$ is a stationary state with the same energy. The proof is straightforward:

$$\hat{H} |\psi'_n\rangle = \hat{H} (\hat{Q} |\psi_n\rangle) = \hat{Q} \hat{H} |\psi_n\rangle = \hat{Q} E_n |\psi_n\rangle = E_n (\hat{Q} |\psi_n\rangle) = E_n |\psi'_n\rangle.$$

For example, if you have an eigenstate of a spherically-symmetric Hamiltonian and you rotate that state about some axis, you must get back another state of the same energy.

You might think that symmetry would *always* lead to degeneracy, and that continuous symmetries would lead to infinite degeneracy, but that is not the case. The reason is that the two states $|\psi_n\rangle$ and $|\psi'_n\rangle$ might be the same.²¹ As an example, consider the Hamiltonian for the harmonic oscillator in one dimension; it commutes with parity. All of its stationary states are either even or odd, so when you act on one with the parity operator you get back the same state

²⁰ When we can't identify the symmetry responsible for a particular degeneracy, we call it an **accidental degeneracy**. In most such cases, the degeneracy turns out to be no accident at all, but instead due to symmetry that is more difficult to identify than, say, rotational invariance. The canonical example is the larger symmetry group of the hydrogen atom (Problem 6.34).

²¹ This is highly non-classical. In classical mechanics, if you take a Keplerian orbit there will always be some axis about which you can rotate it to get a different Keplerian orbit (of the same energy) and in fact there will be an infinite number of such orbits with different orientations. In quantum mechanics, if you rotate the ground state of hydrogen you get back exactly the same state regardless of which axis you choose, and if you rotate one of the states with $n = 2$ and $\ell = 1$, you get back a linear combination of the three orthogonal states with these quantum numbers.

you started with (perhaps multiplied by -1 , but that, physically, *is* the same state). There is therefore no degeneracy associated with inversion symmetry in this case.

In fact, if there is only a single symmetry operator $\hat{Q}$ (or if there are multiple symmetry operators that all commute), you do not get degeneracy in the spectrum. The reason is the same theorem we've now quoted many times: since $\hat{Q}$ and $\hat{H}$ commute, we can find simultaneous eigenstates $|\psi_n\rangle$ of $\hat{Q}$ and $\hat{H}$ and these states transform into themselves under the symmetry operation: $\hat{Q} |\psi_n\rangle = q_n |\psi_n\rangle$.

But what if there are *two* operators that commute with the Hamiltonian (call them $\hat{Q}$ and $\hat{\Lambda}$), but do *not* commute with each other? In this case, degeneracy in the energy spectrum is inevitable. Why?

First, consider a state $|\psi\rangle$ that is an eigenstate of both $\hat{H}$ and $\hat{Q}$ with eigenvalues E_n and q_m respectively. Since $\hat{H}$ and $\hat{\Lambda}$ commute we know that the state $|g\rangle = \hat{\Lambda} |\psi\rangle$ is also an eigenstate of $\hat{H}$ with eigenvalue E_n . Since $\hat{Q}$ and $\hat{\Lambda}$ do *not* commute we know (Section A.5) that there cannot exist a complete set of simultaneous eigenstates of all three operators ($\hat{Q}$, $\hat{\Lambda}$ and $\hat{H}$). Therefore, there must be some $|\psi\rangle$ such that $\hat{\Lambda} |\psi\rangle$ is distinct from $|\psi\rangle$ (specifically, it is not an eigenstate of $\hat{\Lambda}$) meaning that the energy level E_n is at least doubly degenerate. *The presence of multiple non-commuting symmetry operators guarantees degeneracy of the energy spectrum.*

This is precisely the situation we have encountered in the case of central potentials. Here the Hamiltonian commutes with rotations about any axis (or equivalently with the generators $\hat{L}_x$, $\hat{L}_y$, and $\hat{L}_z$) but those rotations don't commute with each other. So we know that there will be degeneracy in the spectrum of a particle in a central potential. The following example shows exactly how much degeneracy is explained by rotational invariance.

Example 6.3

Consider an eigenstate of a central potential $\psi_{n\ell m}$ with energy E_n . Use the fact that the Hamiltonian for a central potential commutes with any component of $\hat{\mathbf{L}}$, and therefore also with $\hat{L}_+$ and $\hat{L}_-$, to show that $\psi_{n\ell m\pm 1}$ are necessarily also eigenstates with the same energy as $\psi_{n\ell m}$.²²

Solution: Since the Hamiltonian commutes with $\hat{L}_\pm$ we have

$$(\hat{H} \hat{L}_\pm - \hat{L}_\pm \hat{H}) \psi_{n\ell m} = 0,$$

so

$$\hat{H} \hat{L}_\pm \psi_{n\ell m} = \hat{L}_\pm \hat{H} \psi_{n\ell m} = E_n \hat{L}_\pm \psi_{n\ell m}$$

or

$$\hat{H} \psi_{n\ell m\pm 1} = E_n \psi_{n\ell m\pm 1}$$

(I canceled the constant $\hbar \sqrt{\ell(\ell+1) - m(m\pm 1)}$ from both sides in the last expression). This argument could obviously be repeated to show that $\psi_{n\ell m\pm 2}$ has the same energy as

²² Of course, we already know the energies are equal since the radial equation, Equation 4.35, does not depend on m . This example demonstrates that rotational invariance is behind the degeneracy.

$\psi_{n\ell m \pm 1}$, and so on until you've exhausted the ladder of states. Therefore, *rotational invariance explains why states which differ only in the quantum number m have the same energy*, and since there are $2\ell + 1$ different values of m , $2\ell + 1$ is the “normal” degeneracy for energies in a central potential.

Of course, the degeneracy of hydrogen (neglecting spin) is n^2 ($= 1, 4, 9, \dots$) (Equation 4.85) which is *greater* than $2\ell + 1$ ($= 1, 3, 5, \dots$).²³ Evidently hydrogen has more degeneracy than is explained by rotational invariance alone. The source of the extra degeneracy is an additional symmetry that is unique to the $1/r$ potential; this is explored in Problem 6.34.²⁴

In this section we have focused on continuous rotational symmetry, but *discrete* rotational symmetry, as experienced (for instance) by an electron in a crystal, can also be of interest. Problem 6.33 explores one such system.

Problem 6.18 Consider the free particle in one dimension: $\hat{H} = \hat{p}^2/2m$. This Hamiltonian has both translational symmetry and inversion symmetry.

(a) Show that translations and inversion don't commute.

(b) Because of the translational symmetry we know that the eigenstates of $\hat{H}$ can be chosen to be simultaneous eigenstates of momentum, namely $f_p(x)$ (Equation 3.32). Show that the parity operator turns $f_p(x)$ into $f_{-p}(x)$; these two states must therefore have the same energy.

(c) Alternatively, because of the inversion symmetry we know that the eigenstates of $\hat{H}$ can be chosen to be simultaneous eigenstates of parity, namely

$$\frac{1}{\sqrt{\pi \hbar}} \cos\left(\frac{p x}{\hbar}\right) \text{ and } \frac{1}{\sqrt{\pi \hbar}} \sin\left(\frac{p x}{\hbar}\right).$$

Show that the translation operator mixes these two states together; they therefore must be degenerate.

Note: Both parity *and* translational invariance are required to explain the degeneracy in the free-particle spectrum. Without parity, there is no reason for $f_p(x)$ and $f_{-p}(x)$ to have the same energy (I mean no reason based on symmetries discussed thus far ... obviously you can plug them in to the time-independent Schrödinger equation and show it's true).

Problem 6.19 For any vector operator $\hat{\mathbf{V}}$ one can define raising and lowering operators as

$$\hat{V}_{\pm} = \hat{V}_x \pm i \hat{V}_y.$$

(a) Using Equation 6.33, show that

$$\begin{aligned} [\hat{L}_z, \hat{V}_{\pm}] &= \pm \hbar \hat{V}_{\pm}. \\ [\hat{L}^2, \hat{V}_{\pm}] &= 2 \hbar^2 \hat{V}_{\pm} \pm 2 \hbar \hat{V}_{\pm} \hat{L}_z \mp 2 \hbar \hat{V}_z \hat{L}_{\pm}. \end{aligned}$$

²³ I don't mean that they necessarily occur in this order. Look back at the infinite spherical well (Figure 4.3): starting with the ground state the degeneracies are $= 1, 3, 5, 1, 7, 3, 9, 5, \dots$. These are precisely the degrees of degeneracy we expect for rotational invariance ($2\ell + 1$ for integer ℓ) but the symmetry considerations don't tell us where in the spectrum each degeneracy will occur.

²⁴ For the three-dimensional harmonic oscillator the degeneracy is $n(n+1)/2 = 1, 3, 6, 10, \dots$ (Problem 4.46) which again is greater than $2\ell + 1$. For a discussion of the additional symmetry in the oscillator problem see D. M. Fradkin, *Am. J. Phys.* **33**, 207 (1965).

(b) Show that, if ψ is an eigenstate of $\hat{L}^2$ and $\hat{L}_z$ with eigenvalues $\ell(\ell+1)\hbar^2$ and $\ell\hbar$ respectively, then either $\hat{V}_+\psi$ is zero or $\hat{V}_+\psi$ is also an eigenstate of $\hat{L}^2$ and $\hat{L}_z$ with eigenvalues $(\ell+1)(\ell+2)\hbar^2$ and $(\ell+1)\hbar$ respectively. This means that, acting on a state with maximal $m_\ell = \ell$, the operator $\hat{V}_+$ either “raises” both the ℓ and m values by 1 or destroys the state.

6.7 ROTATIONAL SELECTION RULES

The most general statement of the rotational selection rules is the **Wigner–Eckart Theorem**; as a practical matter, it is arguably the most important theorem in all of quantum mechanics. Rather than prove the theorem in full generality I will work out the selection rules for the two classes of operators one encounters most often: scalar operators (in Section 6.7.1) and vector operators (in Section 6.7.2). In deriving these selection rules we consider only how the operators behave under a rotation; therefore, the results of this section apply equally well to “true” scalars and pseudoscalars, and those of the next section apply equally well to “true” vectors and pseudeovectors. These selection rules can be combined with the parity selection rules of Section 6.4.3 to obtain a larger set of selection rules for the operator.

6.7.1 Selection Rules for Scalar Operators

The commutation relations for a scalar operator $\hat{f}$ with the three components of angular momentum (Equation 6.34) can be rewritten in terms of the raising and lowering operators as

$$[\hat{L}_z, \hat{f}] = 0 \quad (6.40)$$

$$[\hat{L}_\pm, \hat{f}] = 0 \quad (6.41)$$

$$[\hat{L}^2, \hat{f}] = 0. \quad (6.42)$$

We derive selection rules for $\hat{f}$ by sandwiching these commutators between two states of definite angular momentum, which we will write as $|n \ell m\rangle$ and $|n' \ell' m'\rangle$. These might be hydrogenic orbitals, but they need not be (in fact they need not even be eigenstates of any Hamiltonian but I’ll leave the quantum number n there so they look familiar); we require only that $|n \ell m\rangle$ is an eigenstate of $\hat{L}^2$ and $\hat{L}_z$ with quantum numbers ℓ and m respectively.²⁵

Sandwiching Equation 6.40 between two such states gives

$$\langle n' \ell' m' | [\hat{L}_z, \hat{f}] | n \ell m \rangle = 0$$

or

$$\langle n' \ell' m' | \hat{L}_z \hat{f} | n \ell m \rangle - \langle n' \ell' m' | \hat{f} \hat{L}_z | n \ell m \rangle = 0$$

and therefore

$$(m' - m) \langle n' \ell' m' | \hat{f} | n \ell m \rangle = 0 \quad (6.43)$$

(using the hermiticity of $\hat{L}_z$). Equation 6.43 says that the matrix elements of a scalar operator vanish unless $m' - m \equiv \Delta m = 0$. Repeating this procedure with Equation 6.42 we get

²⁵ Importantly, they satisfy Equations 4.118 and 4.120.

$$\begin{aligned}
\langle n' \ell' m' | [\hat{L}^2, \hat{f}] | n \ell m \rangle &= 0 \\
\langle n' \ell' m' | \hat{L}^2 \hat{f} | n \ell m \rangle - \langle n' \ell' m' | \hat{f} \hat{L}^2 | n \ell m \rangle &= 0 \\
[\ell'(\ell' + 1) - \ell(\ell + 1)] \langle n' \ell' m' | \hat{f} | n \ell m \rangle &= 0.
\end{aligned} \tag{6.44}$$

This tells us that the matrix elements of a scalar operator vanish unless $\ell' - \ell \equiv \Delta\ell = 0$.²⁶ These, then, are the selection rules for a scalar operator: $\Delta\ell = 0$ and $\Delta m = 0$.

However, we can get even more information about the matrix elements from the remaining commutators: (I'll just do the $+$ case and leave the $-$ case for Problem 6.20)

$$\begin{aligned}
\langle n' \ell' m' | [\hat{L}_+, \hat{f}] | n \ell m \rangle &= 0 \\
\langle n' \ell' m' | \hat{L}_+ \hat{f} | n \ell m \rangle - \langle n' \ell' m' | \hat{f} \hat{L}_+ | n \ell m \rangle &= 0 \\
B_{\ell'}^{m'} \langle n' \ell' (m' - 1) | \hat{f} | n \ell m \rangle - A_{\ell}^m \langle n' \ell' m' | \hat{f} | n \ell (m + 1) \rangle &= 0,
\end{aligned} \tag{6.45}$$

where (from Problem 4.21)

$$A_{\ell}^m = \hbar \sqrt{\ell(\ell + 1) - m(m + 1)}, \quad \text{and} \quad B_{\ell}^m = \hbar \sqrt{\ell(\ell + 1) - m(m - 1)}.$$

(I also used the fact that $\hat{L}_{\pm}$ is the Hermitian conjugate of $\hat{L}_{\mp}$: $\langle \psi | \hat{L}_{\pm} = \langle \hat{L}_{\mp} \psi |$.)²⁷ Both terms in Equation 6.45 are zero unless $m' = m + 1$ and $\ell' = \ell$, as we proved in Equations 6.43 and 6.44. When these conditions are satisfied, the two coefficients are equal ($B_{\ell}^{m+1} = A_{\ell}^m$) and Equation 6.45 reduces to

$$\langle n' \ell m | \hat{f} | n \ell m \rangle = \langle n' \ell (m + 1) | \hat{f} | n \ell (m + 1) \rangle. \tag{6.46}$$

Evidently the matrix elements of a scalar operator are independent of m .

The results of this section can be summarized as follows:

$$\langle n' \ell' m' | \hat{f} | n \ell m \rangle = \delta_{\ell\ell'} \delta_{mm'} \langle n' \ell || \hat{f} || n \ell \rangle. \tag{6.47}$$

The funny-looking matrix element on the right, with two bars, is called a **reduced matrix element** and is just shorthand for “a constant that depends on n , ℓ , and n' , but not m .”

Example 6.4

(a) Find $\langle r^2 \rangle$ for all four of the degenerate $n = 2$ states of a hydrogen atom.

Solution: From Equation 6.47 we have, for the states with $\ell = 1$, the following equality:

$$\langle 211 | r^2 | 211 \rangle = \langle 210 | r^2 | 210 \rangle = \langle 21-1 | r^2 | 21-1 \rangle \equiv \langle 21 || r^2 || 21 \rangle.$$

²⁶ The other root of the quadratic $\ell'(\ell' + 1) - \ell(\ell + 1) = 0$ is $\ell' = -(\ell + 1)$; since ℓ and ℓ' are non-negative integers this isn't possible.

²⁷ Since $\hat{L}_x$ and $\hat{L}_y$ are Hermitian,

$$\hat{L}_{\pm}^{\dagger} = (\hat{L}_x \pm i \hat{L}_y)^{\dagger} = \hat{L}_x^{\dagger} \pm (-i) \hat{L}_y^{\dagger} = \hat{L}_x \mp i \hat{L}_y = \hat{L}_{\mp}.$$

To calculate the reduced matrix element we simply pick any one of these expectation values:

$$\begin{aligned}\langle 21 \parallel r^2 \parallel 21 \rangle &= \langle 210 \mid r^2 \mid 210 \rangle \\ &= \int r^2 |\psi_{210}(r)|^2 d^3\mathbf{r} \\ &= \int_0^\infty r^4 |R_{21}(r)|^2 dr \int |Y_1^0(\theta, \phi)|^2 d\Omega.\end{aligned}$$

The spherical harmonics are normalized (Equation 4.31), so the angular integral is 1, and the radial functions $R_{n\ell}$ are listed in Table 4.7, giving

$$\langle 21 \parallel r^2 \parallel 21 \rangle = \int_0^\infty r^4 \frac{1}{24a^3} \frac{r^2}{a^2} e^{-r/a} dr = 30a^2.$$

That determines three of the expectation values. The final expectation value is

$$\begin{aligned}\langle 20 \parallel r^2 \parallel 20 \rangle &= \langle 200 \mid r^2 \mid 200 \rangle \\ &= \int r^2 |\psi_{200}(r)|^2 d^3\mathbf{r} \\ &= \int_0^\infty r^4 |R_{20}(r)|^2 dr \int |Y_0^0(\theta, \phi)|^2 d\Omega \\ &= \int_0^\infty r^4 \frac{1}{2a^3} \left(1 - \frac{1}{2} \frac{r}{a}\right)^2 e^{-r/a} dr \\ &= 42a^2.\end{aligned}$$

Summarizing:

$$\left. \begin{aligned} \langle 200 \mid r^2 \mid 200 \rangle &= 42a^2, \\ \left. \begin{aligned} \langle 211 \mid r^2 \mid 211 \rangle \\ \langle 210 \mid r^2 \mid 210 \rangle \\ \langle 21-1 \mid r^2 \mid 21-1 \rangle \end{aligned} \right\} &= 30a^2. \end{aligned} \right\} \quad (6.48)$$

(b) Find the expectation value of r^2 for an electron in the superposition state

$$|\psi\rangle = \frac{1}{\sqrt{2}} (|200\rangle - i |211\rangle).$$

Solution: We can expand the expectation value as

$$\begin{aligned}\langle \psi \mid r^2 \mid \psi \rangle &= \frac{1}{2} (\langle 200 \mid + i \langle 211 \mid) r^2 (|200\rangle - i |211\rangle) \\ &= \frac{1}{2} (\langle 200 \mid r^2 \mid 200 \rangle + i \langle 211 \mid r^2 \mid 200 \rangle - i \langle 200 \mid r^2 \mid 211 \rangle \\ &\quad + \langle 211 \mid r^2 \mid 211 \rangle).\end{aligned}$$

From Equation 6.47 we see that two of these matrix elements vanish, and

$$\langle \psi \mid r^2 \mid \psi \rangle = \frac{1}{2} (\langle 20 \parallel r^2 \parallel 20 \rangle + \langle 21 \parallel r^2 \parallel 21 \rangle) = 36a^2. \quad (6.49)$$

Problem 6.20 Show that the commutator $[\hat{L}_-, \hat{f}] = 0$ leads to the same rule, Equation 6.46, as does the commutator $[\hat{L}_+, \hat{f}] = 0$.

*

Problem 6.21 For an electron in the hydrogen state

$$\psi = \frac{1}{\sqrt{2}}(\psi_{211} + \psi_{21-1}),$$

find $\langle r \rangle$ after first expressing it in terms of a single reduced matrix element.

6.7.2 Selection Rules for Vector Operators

We now move on to the selection rules for a vector operator $\hat{\mathbf{V}}$. This is significantly more work than the scalar case, but the result is central to understanding atomic transitions (Chapter 11). We begin by defining, by analogy with the angular momentum raising and lowering operators, the operators²⁸

$$\hat{V}_{\pm} \equiv \hat{V}_x \pm i \hat{V}_y.$$

Written in terms of these operators, Equation 6.33 becomes

$$[\hat{L}_z, \hat{V}_z] = 0 \quad (6.50)$$

$$[\hat{L}_z, \hat{V}_{\pm}] = \pm \hbar \hat{V}_{\pm} \quad (6.51)$$

$$[\hat{L}_{\pm}, \hat{V}_{\pm}] = 0 \quad (6.52)$$

$$[\hat{L}_{\pm}, \hat{V}_z] = \mp \hbar \hat{V}_{\pm} \quad (6.53)$$

$$[\hat{L}_{\pm}, \hat{V}_{\mp}] = \pm 2 \hbar \hat{V}_z \quad (6.54)$$

as you will show in Problem 6.22(a).²⁹ Just as for the scalar operator in Section 6.7.1, we sandwich each of these commutators between two states of definite angular momentum to derive (a) conditions under which the matrix elements are guaranteed to vanish and (b) relations between matrix elements with differing values of m or different components of $\hat{\mathbf{V}}$.

From Equation 6.51,

$$\langle n' \ell' m' | \hat{L}_z \hat{V}_{\pm} | n \ell m \rangle - \langle n' \ell' m' | \hat{V}_{\pm} \hat{L}_z | n \ell m \rangle = \pm \hbar \langle n' \ell' m' | \hat{V}_{\pm} | n \ell m \rangle,$$

²⁸ The operators $\hat{V}_{\pm}$ are, up to constants, components of what are known as **spherical tensor** operators of rank 1, written $\hat{T}_q^{(k)}$ where k is the rank and q the component of the operator:

$$\hat{T}_{\pm 1}^{(1)} = \mp \frac{1}{\sqrt{2}} \hat{V}_{\pm} \quad \hat{T}_0^{(1)} = \hat{V}_z.$$

Similarly, the scalar operator f treated in Section 6.7.1 is a rank-0 spherical tensor operator:

$$\hat{T}_0^{(0)} = f.$$

²⁹ Equations 6.51–6.54 each stand for *two* equations: read the upper signs all the way across, or the lower signs.

and since our states are eigenstates of $\hat{L}_z$, this simplifies to

$$[m' - (m \pm 1)] \langle n' \ell' m' | \hat{V}_{\pm} | n \ell m \rangle = 0. \quad (6.55)$$

Equation 6.55 says that *either* $m' = m \pm 1$, or else the matrix element of $\hat{V}_{\pm}$ must vanish. Equation 6.50 works out similarly (see Problem 6.22) and this first set of commutators gives us the selection rules for m :

$$\langle n' \ell' m' | \hat{V}_+ | n \ell m \rangle = 0 \quad \text{unless } m' = m + 1 \quad (6.56)$$

$$\langle n' \ell' m' | \hat{V}_z | n \ell m \rangle = 0 \quad \text{unless } m' = m \quad (6.57)$$

$$\langle n' \ell' m' | \hat{V}_- | n \ell m \rangle = 0 \quad \text{unless } m' = m - 1. \quad (6.58)$$

Note that, if desired, these expressions can be turned back into selection rules for the x - and y -components of our operator, since

$$\begin{aligned} \langle n' \ell' m' | \hat{V}_x | n \ell m \rangle &= \frac{1}{2} \left[\langle n' \ell' m' | \hat{V}_- | n \ell m \rangle + \langle n' \ell' m' | \hat{V}_+ | n \ell m \rangle \right] \\ \langle n' \ell' m' | \hat{V}_y | n \ell m \rangle &= \frac{i}{2} \left[\langle n' \ell' m' | \hat{V}_- | n \ell m \rangle - \langle n' \ell' m' | \hat{V}_+ | n \ell m \rangle \right]. \end{aligned}$$

The remaining commutators, Equations 6.52–6.54, yield a selection rule on ℓ and relations among the nonzero matrix elements. As shown in Problem 6.24, the results may be summarized as³⁰

$$\langle n' \ell' m' | \hat{V}_+ | n \ell m \rangle = -\sqrt{2} C_{m1m'}^{\ell \ 1 \ \ell'} \langle n \ell' \| V \| n \ell \rangle \quad (6.59)$$

$$\langle n' \ell' m' | \hat{V}_- | n \ell m \rangle = \sqrt{2} C_{m-1m'}^{\ell \ 1 \ \ell'} \langle n \ell' \| V \| n \ell \rangle \quad (6.60)$$

$$\langle n' \ell' m' | \hat{V}_z | n \ell m \rangle = C_{m0m'}^{\ell \ 1 \ \ell'} \langle n \ell' \| V \| n \ell \rangle. \quad (6.61)$$

The constants $C_{m_1 m_2 M}^{j_1 j_2 J}$ in these expressions are precisely the Clebsch–Gordan coefficients that appeared in the addition of angular momenta (Section 4.4.3). The Clebsch–Gordan coefficient $C_{m_1 m_2 M}^{j_1 j_2 J}$ vanishes unless $M = m_1 + m_2$ (since the z -components of angular momentum add) and unless $J = j_1 + j_2, j_1 + j_2 - 1, \dots, |j_1 - j_2|$ (Equation 4.182). In particular, the matrix elements of any component of a vector operator, $\langle n' \ell' m' | \hat{V}_i | n \ell m \rangle$, are nonzero only if

$$\Delta \ell = 0, \pm 1, \quad \text{and} \quad \Delta m = 0, \pm 1. \quad (6.62)$$

³⁰ A warning about notation: In the selection rules for the scalar operator r ,

$$\langle n' \ell' m' | r | n \ell m \rangle = \delta_{\ell \ell'} \delta_{m m'} \langle n' \ell' \| r \| n \ell \rangle,$$

and for a component (say z) of the vector operator $\mathbf{r}$,

$$\langle n' \ell' m' | z | n \ell m \rangle = C_{m0m'}^{\ell \ 1 \ \ell'} \langle n' \ell' \| r \| n \ell \rangle,$$

the two reduced matrix elements *are not the same*. One is the reduced matrix element for r and one is the reduced matrix element for $\mathbf{r}$, and these are *different operators* that share the same name. You could tack on a subscript ($\langle n' \ell' \| r \| n \ell \rangle_s$ and $\langle n' \ell' \| r \| n \ell \rangle_v$) to distinguish between the two if that helps keep them straight.

Example 6.5

Find all of the matrix elements of $\hat{\mathbf{r}}$ between the states with $n = 2, \ell = 1$ and $n' = 3, \ell' = 2$:

$$\langle 3 2 m' | r_i | 2 1 m \rangle$$

where $m = -1, 0, 1, m' = -2, -1, 0, 1, 2$, and $r_i = x, y, z$.

Solution: With the vector operator $\hat{\mathbf{V}} = \hat{\mathbf{r}}$, our components are $V_z = z$, $V_+ = x + i y$, and $V_- = x - i y$. We start by calculating *one* of the matrix elements,

$$\begin{aligned} \langle 320 | z | 210 \rangle &= \int \psi_{320}(\mathbf{r}) r \cos \theta \psi_{210}(\mathbf{r}) d^3\mathbf{r} \\ &= \int R_{32}(r)^* r R_{21}(r) r^2 dr \int Y_2^0(\theta, \phi)^* \cos \theta Y_1^0(\theta, \phi) d\Omega \\ &= \frac{2^{12} 3^3 \sqrt{3}}{5^7} a. \end{aligned}$$

From Equation 6.61 we can then determine the reduced matrix element

$$\begin{aligned} \langle 320 | z | 210 \rangle &= C_{000}^{112} \langle 32 \| V \| 21 \rangle \\ \frac{2^{12} 3^3 \sqrt{3}}{5^7} a &= \sqrt{\frac{2}{3}} \langle 32 \| V \| 21 \rangle. \end{aligned}$$

Therefore

$$\langle 32 \| V \| 21 \rangle = \frac{2^{12} 3^4}{5^7 \sqrt{2}} a. \quad (6.63)$$

We can now find all of the remaining matrix elements from Equations 6.59–6.60 with the help of the Clebsch–Gordan table. The relevant coefficients are shown in Figure 6.8. The nonzero matrix elements are

$$\begin{aligned} \langle 322 | \hat{V}_+ | 211 \rangle &= -\sqrt{2} C_{112}^{112} \langle 32 \| V \| 21 \rangle = -\sqrt{2} \langle 32 \| V \| 21 \rangle \\ \langle 321 | \hat{V}_+ | 210 \rangle &= -\sqrt{2} C_{011}^{112} \langle 32 \| V \| 21 \rangle = -\langle 32 \| V \| 21 \rangle \\ \langle 320 | \hat{V}_+ | 21-1 \rangle &= -\sqrt{2} C_{-110}^{112} \langle 32 \| V \| 21 \rangle = -\frac{1}{\sqrt{3}} \langle 32 \| V \| 21 \rangle \\ \langle 320 | \hat{V}_- | 211 \rangle &= \sqrt{2} C_{1-10}^{112} \langle 32 \| V \| 21 \rangle = \frac{1}{\sqrt{3}} \langle 32 \| V \| 21 \rangle \\ \langle 32-1 | \hat{V}_- | 210 \rangle &= \sqrt{2} C_{0-1-1}^{112} \langle 32 \| V \| 21 \rangle = \langle 32 \| V \| 21 \rangle \\ \langle 32-2 | \hat{V}_- | 21-1 \rangle &= \sqrt{2} C_{-1-1-2}^{112} \langle 32 \| V \| 21 \rangle = \sqrt{2} \langle 32 \| V \| 21 \rangle \\ \langle 321 | \hat{V}_z | 211 \rangle &= C_{101}^{112} \langle 32 \| V \| 21 \rangle = \frac{1}{\sqrt{2}} \langle 32 \| V \| 21 \rangle \\ \langle 320 | \hat{V}_z | 210 \rangle &= C_{000}^{112} \langle 32 \| V \| 21 \rangle = \sqrt{\frac{2}{3}} \langle 32 \| V \| 21 \rangle \\ \langle 32-1 | \hat{V}_z | 21-1 \rangle &= C_{-10-1}^{112} \langle 32 \| V \| 21 \rangle = \frac{1}{\sqrt{2}} \langle 32 \| V \| 21 \rangle, \end{aligned}$$

Problem 6.23 The Clebsch–Gordan coefficients are defined by Equation 4.183. Adding together two states with angular momentum j_1 and j_2 produces a state with total angular momentum J according to

$$|J M\rangle = \sum_{m_1, m_2} C_{m_1 m_2 M}^{j_1 j_2 J} |j_1 j_2 m_1 m_2\rangle. \quad (6.64)$$

(a) From Equation 6.64, show that the Clebsch–Gordan coefficients satisfy

$$C_{m_1 m_2 M}^{j_1 j_2 J} = \langle j_1 j_2 m_1 m_2 | J M \rangle. \quad (6.65)$$

(b) Apply $\hat{J}_\pm = \hat{J}_\pm^{(1)} + \hat{J}_\pm^{(2)}$ to Equation 6.64 to derive the **recursion relations for Clebsch–Gordan coefficients**:

$$\begin{aligned} A_J^M C_{m_1 m_2 M+1}^{j_1 j_2 J} &= B_{j_1}^{m_1} C_{m_1-1 m_2 M}^{j_1 j_2 J} + B_{j_2}^{m_2} C_{m_1 m_2-1 M}^{j_1 j_2 J} \\ B_J^M C_{m_1 m_2 M-1}^{j_1 j_2 J} &= A_{j_1}^{m_1} C_{m_1+1 m_2 M}^{j_1 j_2 J} + A_{j_2}^{m_2} C_{m_1 m_2+1 M}^{j_1 j_2 J} \end{aligned} \quad (6.66)$$

Problem 6.24

(a) Sandwich each of the six commutation relations in Equations 6.52–6.54 between $\langle n' \ell' m' |$ and $| n \ell m \rangle$ to obtain relations between matrix elements of $\hat{\mathbf{V}}$. As an example, Equation 6.52 with the upper signs gives

$$B_{\ell'}^{m'} \langle n' \ell' (m' - 1) | V_+ | n \ell m \rangle = A_{\ell}^m \langle n' \ell' m' | V_+ | n \ell (m + 1) \rangle.$$

(b) Using the results in Problem 6.23, show that the six expressions you wrote down in part (a) are satisfied by Equations 6.59–6.61.

*

Problem 6.25 Express the expectation value of the dipole moment $\mathbf{p}_e$ for an electron in the hydrogen state

$$\psi = \frac{1}{\sqrt{2}} (\psi_{211} + \psi_{210})$$

in terms of a single reduced matrix element, and evaluate the expectation value. *Note*: this is the expectation value of a vector so you need to compute all three components. Don't forget Laporte's rule!

6.8 TRANSLATIONS IN TIME

In this section we study time-translation invariance. Consider a solution $\Psi(x, t)$ to the time-dependent Schrödinger equation

$$\hat{H} \Psi(x, t) = i \hbar \frac{\partial}{\partial t} \Psi(x, t).$$

We can define the operator that propagates the wave function forward in time, $\hat{U}(t)$ by

$$\hat{U}(t) \Psi(x, 0) = \Psi(x, t); \quad (6.67)$$

$\hat{U}(t)$ can be expressed in terms of the Hamiltonian, and doing so is straightforward if the Hamiltonian is not itself a function of time. In that case, expanding the right-hand side of Equation 6.67 in a Taylor series gives³³

$$\hat{U}(t) \Psi(x, 0) = \Psi(x, t) = \sum_{n=0}^{\infty} \frac{1}{n!} \frac{\partial^n}{\partial t^n} \Psi(x, t) \Big|_{t=0} t^n \quad (6.68)$$

$$= \sum_{n=0}^{\infty} \frac{1}{n!} \left(-\frac{i}{\hbar} \hat{H} t \right)^n \Psi(x, 0). \quad (6.69)$$

Therefore, in the case of a time-independent Hamiltonian, the time-evolution operator is³⁴

$$\hat{U}(t) = \exp \left[-\frac{i}{\hbar} \hat{H} t \right]. \quad (6.71)$$

We say that the Hamiltonian is the **generator of translations in time**. Note that $\hat{U}(t)$ is a unitary operator (see Problem 6.2).

The time-evolution operator offers a compact way to state the procedure for solving the time-dependent Schrödinger equation. To see the correspondence, write out the wave function at time $t = 0$ as a superposition of stationary states ($\hat{H} \psi_n = E_n \psi_n$):

$$\Psi(x, 0) = \sum_n c_n \psi_n(x).$$

Then

$$\begin{aligned} \Psi(x, t) &= \hat{U}(t) \Psi(x, 0) = \sum_n c_n \hat{U}(t) \psi_n(x) \\ &= \sum_n c_n e^{-i \hat{H} t / \hbar} \psi_n(x) = \sum_n c_n e^{-i E_n t / \hbar} \psi_n(x). \end{aligned}$$

In this sense Equation 6.71 is shorthand for the process of expanding the initial wave function in terms of stationary states and then tacking on the “wobble factors” to obtain the wave function at a later time (Section 2.1).

³³ Why is this analysis limited to the case where $\hat{H}$ is independent of time? Whether or not $\hat{H}$ depends on time, Schrödinger’s equation says $i \hbar \dot{\Psi} = \hat{H} \Psi$. However, if $\hat{H}$ is time dependent then the second derivative of Ψ is given by

$$\frac{\partial^2}{\partial t^2} \Psi = \frac{\partial}{\partial t} \left(\frac{1}{i \hbar} \hat{H} \Psi \right) = \frac{1}{i \hbar} \frac{\partial \hat{H}}{\partial t} \Psi - \frac{1}{\hbar^2} \hat{H}^2 \Psi$$

and higher derivatives will be even more complicated. Therefore, Equation 6.69 only follows from Equation 6.68 when $\hat{H}$ has no time dependence. See also Problem 11.23.

³⁴ This derivation assumes that the actual solution to Schrödinger’s equation, $\Psi(x, t)$, can be expanded as a Taylor series in t , and nothing guarantees that. B. R. Holstein and A. R. Swift, *A. J. Phys.* **40**, 829 (1989) give an innocent-seeming example where such an expansion does not exist. Nonetheless, Equation 6.71 still holds in such cases as long as we define the exponential function through its spectral decomposition (Equation 3.103):

$$\hat{U}(t) \equiv \sum_n e^{-i E_n t / \hbar} |\psi_n\rangle \langle \psi_n|. \quad (6.70)$$

See also M. Amaku et al., *Am. J. Phys.* **85**, 692 (2017).

6.8.1 The Heisenberg Picture

Just as for the other transformations studied in this chapter, we can examine the effect of applying time translation to *operators*, as well as to wave functions. The transformed operators are called **Heisenberg-picture** operators and we follow the convention of giving them a subscript H rather than a prime:

$$\hat{Q}_H(t) = \hat{U}^\dagger(t) \hat{Q} \hat{U}(t). \quad (6.72)$$

Example 6.6

A particle of mass m moves in one dimension in a potential $V(x)$:

$$\hat{H} = \frac{\hat{p}^2}{2m} + V(x).$$

Find the position operator in the Heisenberg picture for an infinitesimal time translation δ .

Solution: From Equation 6.71,

$$\hat{U}(\delta) \approx 1 - i \frac{\delta}{\hbar} \hat{H}.$$

Applying Equation 6.72, we have

$$\begin{aligned} \hat{x}_H(\delta) &\approx \left(1 + i \frac{\delta}{\hbar} \hat{H}^\dagger\right) \hat{x} \left(1 - i \frac{\delta}{\hbar} \hat{H}\right) \\ &\approx \hat{x} - i \frac{\delta}{\hbar} [\hat{x}, \hat{H}] \approx \hat{x} - i \frac{\delta}{\hbar} i \hbar \frac{\hat{p}}{m} \end{aligned}$$

so

$$\hat{x}_H(\delta) \approx \hat{x}_H(0) + \frac{1}{m} \hat{p}_H(0) \delta$$

(making use of the fact that the Heisenberg-picture operators at time 0 are just the untransformed operators). This looks exactly like classical mechanics: $x(\delta) \approx x(0) + v(0) \delta$. The Heisenberg picture illuminates the connection between classical and quantum mechanics: the quantum operators obey the classical equations of motion (see Problem 6.29).

Example 6.7

A particle of mass m moves in one dimension in a harmonic-oscillator potential:

$$\hat{H} = \frac{\hat{p}^2}{2m} + \frac{1}{2} m \omega^2 x^2.$$

Find the position operator in the Heisenberg picture at time t .

Solution: Consider the action of $\hat{x}_H$ on a stationary state ψ_n . (Introducing ψ_n allows us to replace the *operator* $e^{-i \hat{H} t / \hbar}$ with the *number* $e^{-i E_n t / \hbar}$, since $e^{-i \hat{H} t / \hbar} \psi_n = e^{-i E_n t / \hbar} \psi_n$.) Writing $\hat{x}$ in terms of raising and lowering operators we have (using Equations 2.62, 2.67, and 2.70)

$$\begin{aligned} \hat{x}_H(t) \psi_n(x) &= \hat{U}^\dagger(t) \hat{x} \hat{U}(t) \psi_n(x) \\ &= e^{i \hat{H} t / \hbar} \sqrt{\frac{\hbar}{2m\omega}} (\hat{a}_+ + \hat{a}_-) e^{-i \hat{H} t / \hbar} \psi_n(x) \end{aligned}$$

$$\begin{aligned}
&= \sqrt{\frac{\hbar}{2m\omega}} e^{-i E_n t/\hbar} e^{i \hat{H} t/\hbar} (\hat{a}_+ + \hat{a}_-) \psi_n(x) \\
&= \sqrt{\frac{\hbar}{2m\omega}} e^{-i E_n t/\hbar} e^{i \hat{H} t/\hbar} \left[\sqrt{n+1} \psi_{n+1}(x) + \sqrt{n} \psi_{n-1}(x) \right] \\
&= \sqrt{\frac{\hbar}{2m\omega}} e^{-i E_n t/\hbar} \left[\sqrt{n+1} e^{i E_{n+1} t/\hbar} \psi_{n+1}(x) \right. \\
&\quad \left. + \sqrt{n} e^{i E_{n-1} t/\hbar} \psi_{n-1}(x) \right] \\
&= \sqrt{\frac{\hbar}{2m\omega}} \left[\sqrt{n+1} e^{i \omega t} \psi_{n+1}(x) + \sqrt{n} e^{-i \omega t} \psi_{n-1}(x) \right]. \quad (6.73)
\end{aligned}$$

Thus³⁵

$$\hat{x}_H(t) = \sqrt{\frac{\hbar}{2m\omega}} \left[e^{i \omega t} \hat{a}_+ + e^{-i \omega t} \hat{a}_- \right].$$

Or, using Equation 2.48 to express $\hat{a}_{\pm}$ in terms of $\hat{x}$ and $\hat{p}$,

$$\hat{x}_H(t) = \hat{x}_H(0) \cos(\omega t) + \frac{1}{m\omega} \hat{p}_H(0) \sin(\omega t). \quad (6.74)$$

As in Example 6.6 we see that the Heisenberg-picture operator satisfies the *classical* equation of motion for a mass on a spring.

In this book we have been working in the **Schrödinger picture**, so-named by Dirac because it was the picture that Schrödinger himself had in mind. In the Schrödinger picture, the wave function evolves in time according to the Schrödinger equation

$$\hat{H} \Psi(x, t) = i \hbar \frac{\partial}{\partial t} \Psi(x, t).$$

The operators $\hat{x} = x$ and $\hat{p} = -i \hbar \partial_x$ have no time dependence of their own, and the time dependence of expectation values (or, more generally, matrix elements) comes from the time dependence of the wave function:³⁶

$$\langle \hat{Q} \rangle = \langle \Psi(t) | \hat{Q} | \Psi(t) \rangle.$$

In the Heisenberg picture, the wave function is constant in time, $\Psi_H(x) = \Psi(x, 0)$, and the *operators* evolve in time according to Equation 6.72. In the Heisenberg picture, the time dependence of expectation values (or matrix elements) is carried by the *operators*.

$$\langle \hat{Q} \rangle = \langle \Psi_H | \hat{Q}_H(t) | \Psi_H \rangle.$$

Of course, the two pictures are entirely equivalent since

$$\langle \Psi(t) | \hat{Q} | \Psi(t) \rangle = \langle \Psi(0) | \hat{U}^\dagger \hat{Q} \hat{U} | \Psi(0) \rangle = \langle \Psi_H | \hat{Q}_H(t) | \Psi_H \rangle.$$

³⁵ Since Equation 6.73 holds for *any* stationary state ψ_n and since the ψ_n constitute a complete set of states, the operators must in fact be identical.

³⁶ I am assuming that $\hat{Q}$, like $\hat{x}$ or $\hat{p}$, has no *explicit* time dependence.

A nice analogy for the two pictures runs as follows. On an ordinary clock, the hands move in a clockwise direction while the numbers stay fixed. But one could equally well design a clock where the hands are stationary and the numbers move in the counter-clockwise direction. The correspondence between these two clocks is roughly the correspondence between the Schrödinger and Heisenberg pictures, the hands representing the wave function and the numbers representing the operator. Other pictures could be introduced as well, in which both the hands of the clock and the numbers on the dial move at intermediate rates such that the clock still tells the correct time.³⁷

- * **Problem 6.26** Work out $\hat{p}_H(t)$ for the system in Example 6.7 and comment on the correspondence with the classical equation of motion.

- ** **Problem 6.27** Consider a free particle of mass m . Show that the position and momentum operators in the Heisenberg picture are given by

$$\begin{aligned}\hat{x}_H(t) &= \hat{x}_H(0) + \frac{1}{m} \hat{p}_H(0) t \\ \hat{p}_H(t) &= \hat{p}_H(0).\end{aligned}$$

Comment on the relationship between these equations and the classical equations of motion. *Hint:* you will first need to evaluate the commutator $[\hat{x}, \hat{H}^n]$; this will allow you to evaluate the commutator $[\hat{x}, \hat{U}]$.

6.8.2 Time-Translation Invariance

If the Hamiltonian is time-dependent one can still write the formal solution to the Schrödinger equation in terms of the time-translation operator, $\hat{U}$:

$$\Psi(x, t) = \hat{U}(t, t_0) \Psi(x, t_0), \quad (6.75)$$

but $\hat{U}$ no longer takes the simple form 6.71.³⁸ (See Problem 11.23 for the general case.) For an infinitesimal time interval δ (see Problem 6.28)

$$\hat{U}(t_0 + \delta, t_0) \approx 1 - \frac{i}{\hbar} \hat{H}(t_0) \delta. \quad (6.76)$$

Time-translation invariance means that the time evolution is independent of which time interval we are considering. In other words

$$\hat{U}(t_1 + \delta, t_1) = \hat{U}(t_2 + \delta, t_2) \quad (6.77)$$

for any choice of t_1 and t_2 . This ensures that if the system starts in state $|\alpha\rangle$ at time t_1 and evolves for a time δ then it will end up in the same state $|\beta\rangle$ as if the system started in the same state $|\alpha\rangle$ at time t_2 and evolved for the same amount of time δ ; i.e. the experiment proceeds the same on Thursday as it did on Tuesday, assuming identical conditions. Plugging Equation 6.76 into Equation 6.77 we see that the requirement for this to be true is $\hat{H}(t_1) = \hat{H}(t_2)$, and since

³⁷ Of these other possible pictures the most important is the **interaction picture** (or Dirac picture) which is often employed in time-dependent perturbation theory.

³⁸ And is a function of *both* the initial time t_0 and the final time t , not simply the amount of time for which the wave function has evolved.

this must hold true for all t_1 and t_2 , it must be that the Hamiltonian is in fact time-independent after all (for time-translation invariance to hold):

$$\frac{\partial \hat{H}}{\partial t} = 0.$$

In that case the generalized Ehrenfest theorem says

$$\frac{d}{dt} \langle \hat{H} \rangle = \frac{i}{\hbar} \langle [\hat{H}, \hat{H}] \rangle + \left\langle \frac{\partial \hat{H}}{\partial t} \right\rangle = 0.$$

Therefore, **energy conservation is a consequence of time-translation invariance.**

We have now recovered all the classical conservation laws: conservation of momentum, energy, and angular momentum, and seen that they are each related to a continuous symmetry of the Hamiltonian (spatial translation, time translation, and rotation, respectively). And in quantum mechanics, discrete symmetries (such as parity) can also lead to conservation laws.

Problem 6.28 Show that Equations 6.75 and 6.76 are the solution to the Schrödinger equation for an infinitesimal time δ . *Hint:* expand $\Psi(x, t)$ in a Taylor series.

* **Problem 6.29** Differentiate Equation 6.72 to obtain the **Heisenberg equations of motion**

$$i \hbar \frac{d}{dt} \hat{Q}_H(t) = [\hat{Q}_H(t), \hat{H}] \quad (6.78)$$

(for $\hat{Q}$ and $\hat{H}$ independent of time).³⁹ Plug in $\hat{Q} = \hat{x}$ and $\hat{Q} = \hat{p}$ to obtain the differential equations for $\hat{x}_H$ and $\hat{p}_H$ in the Heisenberg picture for a single particle of mass m moving in a potential $V(x)$.

*** **Problem 6.30** Consider a time-independent Hamiltonian for a particle moving in one dimension that has stationary states $\psi_n(x)$ with energies E_n .

(a) Show that the solution to the time-dependent Schrödinger equation can be written

$$\Psi(x, t) = \hat{U}(t) \Psi(x, 0) = \int K(x, x', t) \Psi(x', 0) dx',$$

where $K(x, x', t)$, known as the **propagator**, is

$$K(x, x', t) = \sum_n \psi_n^*(x') e^{-i E_n t / \hbar} \psi_n(x). \quad (6.79)$$

Here $|K(x, x', t)|^2$ is the probability for a quantum mechanical particle to travel from position x' to position x in time t .

(b) Find K for a particle of mass m in a simple harmonic oscillator potential of frequency ω . You will need the identity

³⁹ For time-dependent $\hat{Q}$ and $\hat{H}$ the generalization is

$$i \hbar \frac{d}{dt} \hat{Q}_H(t) = [\hat{Q}_H(t), \hat{H}_H(t)] + \hat{U}^\dagger \frac{\partial \hat{Q}}{\partial t} \hat{U}.$$

$$\frac{1}{\sqrt{1-z^2}} \exp \left[-\frac{\xi^2 + \eta^2 - 2\xi\eta z}{1-z^2} \right] = e^{-\xi^2} e^{-\eta^2} \sum_{n=0}^{\infty} \frac{z^n}{2^n n!} H_n(\xi) H_n(\eta).$$

- (c) Find $\Psi(x, t)$ if the particle from part (a) is initially in the state⁴⁰

$$\Psi(x, 0) = \left(\frac{2a}{\pi} \right)^{1/4} e^{-a(x-x_0)^2}.$$

Compare your answer with Problem 2.49. *Note:* Problem 2.49 is a special case with $a = m\omega/2\hbar$.

- (d) Find K for a free particle of mass m . In this case the stationary states are continuous, not discrete, and one must make the replacement

$$\sum_n \rightarrow \int_{-\infty}^{\infty} dp$$

in Equation 6.79.

- (e) Find $\Psi(x, t)$ for a free particle that starts out in the state

$$\Psi(x, 0) = \left(\frac{2a}{\pi} \right)^{1/4} e^{-a x^2}.$$

Compare your answer with Problem 2.21.

FURTHER PROBLEMS ON CHAPTER 6

Problem 6.31 In deriving Equation 6.3 we assumed that our function had a Taylor series. The result holds more generally if we define the exponential of an operator by its spectral decomposition,

$$\hat{T}(a) = \int e^{-i a p/\hbar} |p\rangle \langle p| dp, \quad (6.80)$$

rather than its power series. Here I've given the operator in Dirac notation; acting on a position-space function (see the discussion on page 123) this means

$$\hat{T}(a) \psi(x) = \int_{-\infty}^{\infty} e^{-i a p/\hbar} f_p(x) \Phi(p) dp, \quad (6.81)$$

where $\Phi(p)$ is the momentum space wave function corresponding to $\psi(x)$ and $f_p(x)$ is defined in Equation 3.32. Show that the operator $\hat{T}(a)$, as given by Equation 6.81, applied to the function

$$\psi(x) = \sqrt{\lambda} e^{-\lambda|x|}$$

(whose first derivative is undefined at $x = 0$) gives the correct result.

⁴⁰ The integrals in (c)–(e) can *all* be done with the following identity:

$$\int_{-\infty}^{\infty} e^{-ax^2+bx} dx = \sqrt{\frac{\pi}{a}} e^{b^2/4a},$$

which was derived in Problem 2.21.

**

Problem 6.32 Rotations on spin states are given by an expression identical to Equation 6.32, with the spin angular momentum replacing the orbital angular momentum:

$$R_{\mathbf{n}}(\varphi) = \exp \left[-i \frac{\varphi}{\hbar} \mathbf{n} \cdot \mathbf{S} \right].$$

In this problem we will consider rotations of a spin-1/2 state.

(a) Show that

$$(\mathbf{a} \cdot \boldsymbol{\sigma})(\mathbf{b} \cdot \boldsymbol{\sigma}) = \mathbf{a} \cdot \mathbf{b} + i(\mathbf{a} \times \mathbf{b}) \cdot \boldsymbol{\sigma},$$

where the σ_i are the Pauli spin matrices and $\mathbf{a}$ and $\mathbf{b}$ are ordinary vectors. Use the result of Problem 4.29.

(b) Use your result from part (a) to show that

$$\exp \left[-i \frac{\varphi}{\hbar} \mathbf{n} \cdot \mathbf{S} \right] = \cos \left(\frac{\varphi}{2} \right) - i \sin \left(\frac{\varphi}{2} \right) \mathbf{n} \cdot \boldsymbol{\sigma}.$$

Recall that $\mathbf{S} = (\hbar/2) \boldsymbol{\sigma}$.

(c) Show that your result from part (b) becomes, in the standard basis of spin up and spin down along the z axis, the matrix

$$R_{\mathbf{n}} = \cos \left(\frac{\varphi}{2} \right) \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} - i \sin \left(\frac{\varphi}{2} \right) \begin{pmatrix} \cos \theta & \sin \theta e^{-i\phi} \\ \sin \theta e^{i\phi} & -\cos \theta \end{pmatrix}$$

where θ and ϕ are the polar coordinates of the unit vector $\mathbf{n}$ that describes the axis of rotation.

(d) Verify that the matrix $R_{\mathbf{n}}$ in part (c) is unitary.

(e) Compute explicitly the matrix $S'_x = R^\dagger S_x R$ where R is a rotation by an angle φ about the z axis and verify that it returns the expected result. *Hint:* rewrite your result for S'_x in terms of S_x and S_y .

(f) Construct the matrix for a π rotation about the x axis and verify that it turns an up spin into a down spin.

(g) Find the matrix describing a 2π rotation about the z axis. Why is this answer surprising?⁴¹



Problem 6.33 Consider a particle of mass m in a two-dimensional infinite square well with sides of length L . With the origin placed at the center of the well, the stationary states can be written as

$$\psi_{n_x n_y}(x, y) = \frac{2}{L} \sin \left[\frac{n_x \pi}{L} \left(x - \frac{L}{2} \right) \right] \sin \left[\frac{n_y \pi}{L} \left(y - \frac{L}{2} \right) \right],$$

with energies

$$E_{n_x n_y} = \frac{\pi^2 \hbar^2}{2m L^2} (n_x^2 + n_y^2),$$

for positive integers n_x and n_y .

⁴¹ For a discussion of how this sign change is actually *measured*, see S. A. Werner et al., *Phys. Rev. Lett.* **35**, 1053 (1975).

- (a) The two states ψ_{ab} and ψ_{ba} for $a \neq b$ are clearly degenerate. Show that a rotation by 90° counterclockwise about the center of the square carries one into the other,

$$\hat{R} \psi_{ab} \propto \psi_{ba},$$

and determine the constant of proportionality. *Hint:* write ψ_{ab} in polar coordinates.

- (b) Suppose that instead of ψ_{ab} and ψ_{ba} we choose the basis ψ_+ and ψ_- for our two degenerate states:

$$\psi_{\pm} = \frac{\psi_{ab} \pm \psi_{ba}}{\sqrt{2}}.$$

Show that if a and b are both even or both odd, then ψ_+ and ψ_- are eigenstates of the rotation operator.

- (c) Make a contour plot of the state ψ_- for $a = 5$ and $b = 7$ and verify (visually) that it is an eigenstate of every symmetry operation of the square (rotation by an integer multiple of $\pi/2$, reflection across a diagonal, or reflection along a line bisecting two sides). The fact that ψ_+ and ψ_- are not connected to each other by any symmetry of the square means that there must be additional symmetry explaining the degeneracy of these two states.⁴²

Problem 6.34 The Coulomb potential has *more symmetry* than simply rotational invariance. This additional symmetry is manifest in an additional conserved quantity, the **Laplace–Runge–Lenz vector**

$$\hat{\mathbf{M}} = \frac{\hat{\mathbf{p}} \times \hat{\mathbf{L}} - \hat{\mathbf{L}} \times \hat{\mathbf{p}}}{2m} + V(r) \mathbf{r},$$

where $V(\mathbf{r})$ is the potential energy, $V(r) = -e^2/4\pi\epsilon_0 r$.⁴³ The complete set of commutators for the conserved quantities in the hydrogen atom is

$$\begin{aligned} \text{(i)} \quad & [\hat{H}, \hat{M}_i] = 0 \\ \text{(ii)} \quad & [\hat{H}, \hat{L}_i] = 0 \\ \text{(iii)} \quad & [\hat{L}_i, \hat{L}_j] = i\hbar\epsilon_{ijk}\hat{L}_k \\ \text{(iv)} \quad & [\hat{L}_i, \hat{M}_j] = i\hbar\epsilon_{ijk}\hat{M}_k \\ \text{(v)} \quad & [\hat{M}_i, \hat{M}_j] = \frac{\hbar}{i}\epsilon_{ijk}L_k\frac{2}{m}\hat{H}. \end{aligned}$$

The *physical* content of these equations is that (i) $\mathbf{M}$ is a conserved quantity, (ii) $\mathbf{L}$ is a conserved quantity, (iii) $\mathbf{L}$ is a vector, and (iv) $\mathbf{M}$ is a vector ((v) has no obvious interpretation). There are two additional relations between the quantities $\hat{\mathbf{L}}$, $\hat{\mathbf{M}}$, and $\hat{H}$. They are

⁴² See F. Leyvraz, et al., *Am. J. Phys.* **65**, 1087 (1997) for a discussion of this “accidental” degeneracy.

⁴³ The full symmetry of the Coulomb Hamiltonian is not just the obvious three-dimensional rotation group (known to mathematicians as $SO(3)$), but the four-dimensional rotation group ($SO(4)$), which has six generators ($\mathbf{L}$ and $\mathbf{M}$). (If the four axes are w , x , y , and z , the generators correspond to rotations in each of the six orthogonal planes, wx , wy , wz (that’s $\mathbf{M}$) and yz , zx , xy (that’s $\mathbf{L}$)).

$$(vi) \quad \hat{M}^2 = \left(\frac{e^2}{4\pi\epsilon_0} \right)^2 + \frac{2}{m} \hat{H} (\hat{L}^2 + \hbar^2)$$

$$(vii) \quad \hat{\mathbf{M}} \cdot \hat{\mathbf{L}} = 0.$$

(a) From the result of Problem 6.19, and the fact that $\hat{\mathbf{M}}$ is a conserved quantity, we know that $\hat{M}_+ \psi_{n\ell\ell} = c_{n\ell} \psi_{n(\ell+1)(\ell+1)}$ for some constants $c_{n\ell}$. Apply (vii) to the state $\psi_{n\ell\ell}$ to show that

$$\hat{M}_z \psi_{n\ell\ell} = -\frac{1}{\sqrt{2}} \frac{1}{\sqrt{\ell+1}} c_{n\ell} \psi_{n(\ell+1)\ell}.$$

(b) Use (vi) to show that

$$\hat{M}_- \hat{M}_+ \psi_{n\ell\ell} = \left(\frac{e^2}{4\pi\epsilon_0} \right)^2 \left[1 - \left(\frac{\ell+1}{n} \right)^2 \right] \psi_{n\ell\ell} - \hat{M}_z^2 \psi_{n\ell\ell}.$$

(c) From your results to parts (a) and (b), obtain the constants $c_{n\ell}$. You should find that $c_{n\ell}$ is nonzero unless $\ell = n - 1$. *Hint:* Consider $\int |M_+ \psi_{n\ell m}|^2 d^3\mathbf{r}$ and use the fact that $M_{\pm}$ are Hermitian conjugates. Figure 6.9 shows how the degenerate states of hydrogen are related by the generators $\hat{\mathbf{L}}$ and $\hat{\mathbf{M}}$.

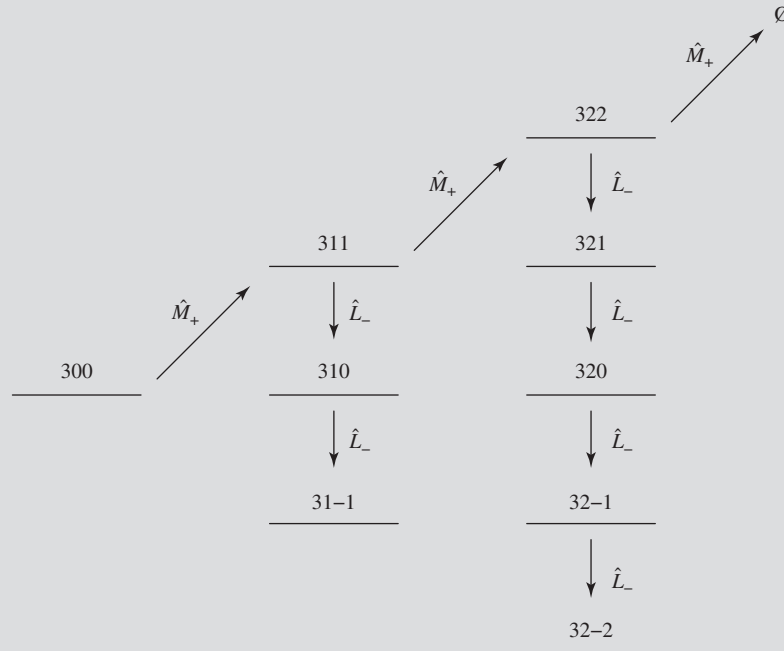


Figure 6.9: The degenerate $n = 3$ states of the hydrogen atom, and the symmetry operations that connect them.

**

Problem 6.35 A Galilean transformation performs a boost from a reference frame $\mathcal{S}$ to a reference frame $\mathcal{S}'$ moving with velocity $-v$ with respect to $\mathcal{S}$ (the origins of the two frames coincide at $t = 0$). The unitary operator that carries out a Galilean transformation at time t is

$$\hat{\Gamma}(v, t) = \exp \left[-\frac{i}{\hbar} v (t \hat{p} - m \hat{x}) \right].$$

- (a) Find $\hat{x}' = \hat{\Gamma}^\dagger \hat{x} \hat{\Gamma}$ and $\hat{p}' = \hat{\Gamma}^\dagger \hat{p} \hat{\Gamma}$ for an infinitesimal transformation with velocity δ . What is the physical meaning of your result?
 (b) Show that

$$\begin{aligned} \hat{\Gamma}(v, t) &= \exp \left[\frac{i}{\hbar} \left(m x v - \frac{1}{2} m v^2 t \right) \right] \hat{T}(vt) \\ &= \hat{T}(vt) \exp \left[\frac{i}{\hbar} \left(m x v + \frac{1}{2} m v^2 t \right) \right]. \end{aligned}$$

where $\hat{T}$ is the spatial translation operator (Equation 6.3). You will need to use the Baker–Campbell–Hausdorff formula (Problem 3.29).

- (c) Show that if Ψ is a solution to the time-dependent Schrödinger equation with Hamiltonian

$$\hat{H} = \frac{\hat{p}^2}{2m} + V(x)$$

then the boosted wave function $\Psi' = \hat{\Gamma}(v, t) \Psi$ is a solution to the time-dependent Schrödinger equation with the potential $V(x)$ in motion:

$$\hat{H} = \frac{\hat{p}^2}{2m} + V(x - vt).$$

Note: $(d/dt) e^{\hat{A}} = e^{\hat{A}} (d\hat{A}/dt)$ only if $[\hat{A}, (d\hat{A}/dt)] = 0$.

- (d) Show that the result of Problem 2.50(a) is an example of this result.

Problem 6.36 A ball thrown through the air leaves your hand at position $\mathbf{r}_0$ with a velocity of $\mathbf{v}_0$ and arrives a time t later at position $\mathbf{r}_1$ traveling with a velocity $\mathbf{v}_1$ (Figure 6.10). Suppose we could instantaneously reverse the ball's velocity when it reaches $\mathbf{r}_1$. Neglecting air resistance, it would retrace the path that took it from $\mathbf{r}_0$ to $\mathbf{r}_1$ and arrive back at $\mathbf{r}_0$ after another time t had passed, traveling with a velocity $-\mathbf{v}_0$. This is an example of **time-reversal invariance**—reverse the motion of a particle at any point along its trajectory and it will retrace its path with an equal and opposite velocity at all positions.

Why is this called time reversal? After all, it was the velocity that was reversed, *not* time. Well, if we showed you a movie of the ball traveling from $\mathbf{r}_1$ to $\mathbf{r}_0$, there would be no way to tell if you were watching a movie of the ball after the reversal playing forward, or a movie of the ball before the reversal playing backward. In a time-reversal invariant system, playing the movie backwards represents another possible motion.

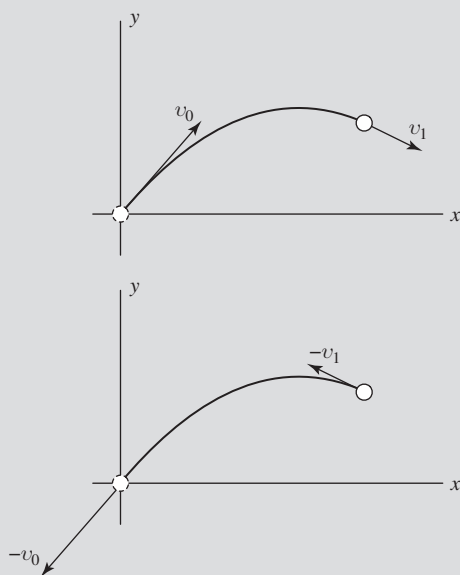


Figure 6.10: A ball thrown through the air (ignore air resistance) is an example of a system with time-reversal symmetry. If we flip the velocity of the particle at any point along its trajectory, it will retrace its path.

A familiar example of a system that does *not* exhibit time-reversal symmetry is a charged particle moving in an external magnetic field.⁴⁴ In that case, when you reverse the velocity of the particle, the Lorentz force will also change sign and the particle will not retrace its path; this is illustrated in Figure 6.11.

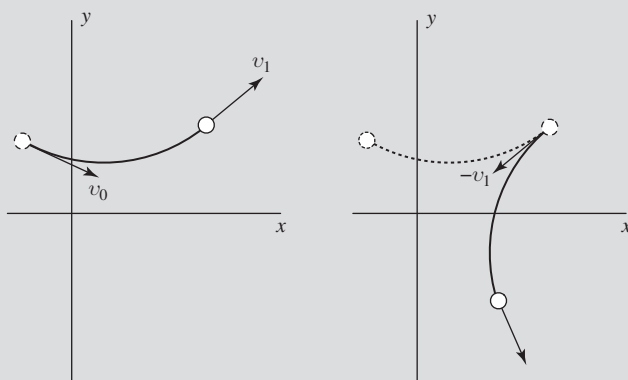


Figure 6.11: An external magnetic field breaks time-reversal symmetry. Shown is the trajectory of a particle of charge $+q$ traveling in a uniform magnetic field pointing into the page. If we flip the particle's velocity from $\mathbf{v}_1$ to $-\mathbf{v}_1$ at the point shown, the particle does not retrace its path, but instead moves onto a new circular orbit.

⁴⁴ By external magnetic field, I mean that we only reverse the velocity of our charge q , and not the velocities of the charges producing the magnetic field. If we reversed those velocities as well, the magnetic field would also switch directions, the Lorentz force on the charge q would be unchanged by the reversal, and the system would in fact be time-reversal invariant.

The **time-reversal operator** $\hat{\Theta}$ is the operator that reverses the momentum of the particle ($\mathbf{p} \rightarrow -\mathbf{p}$), leaving its position unchanged. A better name would really be the “reversal of the direction of motion” operator.⁴⁵ For a spinless particle, the time-reversal operator $\hat{\Theta}$ simply complex conjugates the position-space wave function⁴⁶

$$\hat{\Theta} \Psi(x, t) = \Psi^*(x, t). \quad (6.82)$$

(a) Show that the operators $\hat{x}$ and $\hat{p}$ transform under time reversal as

$$\begin{aligned} \hat{x}' &= \hat{\Theta}^{-1} \hat{x} \hat{\Theta} = \hat{x} \\ \hat{p}' &= \hat{\Theta}^{-1} \hat{p} \hat{\Theta} = -\hat{p}. \end{aligned}$$

Hint: Do this by calculating the action of $\hat{x}'$ and $\hat{p}'$ on an arbitrary test function $f(x)$.

(b) We can write down a mathematical statement of time-reversal invariance from our discussion above. We take a system, evolve it for a time t , reverse its momentum, and evolve it for time t again. If the system is time-reversal invariant it will be back where it started, albeit with its momentum reversed (Figure 6.10). As an operator statement this says

$$\hat{U}(t) \hat{\Theta} \hat{U}(t) = \hat{\Theta}.$$

If this is to hold for any time interval, it must hold in particular for an infinitesimal time interval δ . Show that time-reversal invariance requires

$$[\hat{\Theta}, \hat{H}] = 0. \quad (6.83)$$

(c) Show that, for a time-reversal invariant Hamiltonian, if $\psi_n(x)$ is a stationary state with energy E_n , then $\psi_n^*(x)$ is also a stationary state with the same energy E_n . If the energy is nondegenerate, this means that the stationary state can be chosen as real.

(d) What do you get by time-reversing a momentum eigenfunction $f_p(x)$ (Equation 3.32)? How about a hydrogen wave function $\psi_{n\ell m}(r, \theta, \phi)$? Comment on each state's relation to the untransformed state and verify that the transformed and untransformed states share the same energy, as guaranteed by (c).

Problem 6.37 As an angular momentum, a particle's spin must flip under time reversal (Problem 6.36). The action of time-reversal on a spinor (Section 4.4.1) is in fact

⁴⁵ See Eugene P. Wigner, *Group Theory and its Applications to Quantum Mechanics and Atomic Spectra* (Academic Press, New York, 1959), p. 325.

⁴⁶ Time reversal is an **anti-unitary operator**. An anti-unitary operator satisfies

$$\begin{aligned} \langle \Theta f | \Theta g \rangle &= \langle f | g \rangle^* \\ \hat{\Theta} (a |\alpha\rangle + b |\beta\rangle) &= a^* \hat{\Theta} |\alpha\rangle + b^* \hat{\Theta} |\beta\rangle \end{aligned}$$

whereas a unitary operator satisfies the same two equations without the complex conjugates. I won't define the adjoint of an anti-unitary operator; instead I use $\hat{\Theta}^{-1}$ for an anti-unitary operator where we might have used $\hat{U}^\dagger$ or $\hat{U}^{-1}$ interchangeably for a unitary operator.

$$\hat{\Theta} \begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} -b^* \\ a^* \end{pmatrix} \quad (6.84)$$

so that, in addition to the complex conjugation, the up and down components are interchanged.⁴⁷

(a) Show that $\hat{\Theta}^2 = -1$ for a spin-1/2 particle.

(b) Consider an eigenstate $|\psi_n\rangle$ of a time-reversal invariant Hamiltonian (Equation 6.83) with energy E_n . We know that $|\psi'_n\rangle = \hat{\Theta} |\psi_n\rangle$ is also an eigenstate of $\hat{H}$ with the same energy E_n . There two possibilities: either $|\psi'_n\rangle$ and $|\psi_n\rangle$ are the *same* state (meaning $|\psi'_n\rangle = c |\psi_n\rangle$ for some complex constant c) or they are distinct states. Show that the first case leads to a contradiction in the case of a spin-1/2 particle, meaning the energy level must be (at least) two-fold degenerate in that case.

Comment: What you have proved is a special case of **Kramer's degeneracy**: for an odd number of spin-1/2 particles (or any half-integer spin for that matter), every energy level (of a time-reversal-invariant Hamiltonian) is at least two-fold degenerate. This is because—as you just showed—for half-integer spin a state and its time-reversed state are necessarily distinct.⁴⁸

⁴⁷ For arbitrary spin,

$$\hat{\Theta} = e^{-i\pi \hat{S}_y/\hbar} \hat{K} \quad (6.85)$$

where the first term is a rotation by π about the y axis and $\hat{K}$ is the operator that performs the complex conjugation.

⁴⁸ What about in the case of a spin-0 particle—does time-reversal symmetry tell us anything interesting? Actually it does. For one thing, the stationary states can be chosen as real; you proved this back in Problem 2.2 but we now see that it is a consequence of time-reversal symmetry. Another example is the degeneracy of the energy levels in a periodical potential (Section 5.3.2 and Problem 6.6) for states with crystal momentum q and $-q$. This can be ascribed to inversion symmetry if the potential is symmetric, but the degeneracy persists even when inversion symmetry is absent (try it out!); that is a result of time-reversal symmetry.